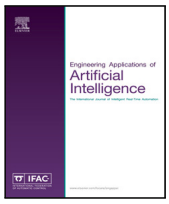
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Survey paper



Recent trends of retrieval-augmented generation in healthcare: A systematic review

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ABSTRACT

The rapid development of large language models has opened new possibilities in healthcare, assisting practitioners, and making informed decisions. However, these models face significant challenges such as hallucination and outdated knowledge, particularly in specialized medical domains where accurate and timely information retrieval is critical. Retrieval-augmented generation has emerged as a promising solution that incorporates external, up-to-date medical knowledge to enhance language model outputs. In this paper, we present a comprehensive survey of such applications across diverse healthcare fields, analyzing their retrieval mechanisms and benefits. We systematically review high-quality studies published between 2020-2025, covering general medicine, biomedical research, mental health, and other specialties, highlighting domain-specific challenges and solutions. While domains such as oncology and radiology show growing adoption of retrieval-augmented generation, significant gaps persist in high-priority areas like neurology that remain largely underexplored, revealing important limitations in current development and deployment. A key finding reveals that most implementations rely on basic architectures, underutilizing advanced pre- and post-retrieval techniques essential for clinical accuracy. To our knowledge, this is the first survey to provide a detailed, field-by-field analysis of retrieval-augmented generation in healthcare, offering insights into current limitations and future directions for this transformative technology.

1. Introduction

Recent advances in deep learning have revolutionized clinical language understanding, offering unprecedented opportunities for enhancing healthcare delivery (Shi et al., 2022; Landolsi et al., 2024; Li et al., 2021). The emergence of Large Language Models (LLMs) as ChatGPT has particularly catalyzed this transformation, enabling intelligent systems that support clinical decision-making, streamline diagnostics, and improve patient care quality (Stafie et al., 2023; Goldberg et al., 2024; Wachter and Brynjolfsson, 2024). These generative AI systems demonstrate remarkable capabilities in processing medical knowledge, interpreting complex patient records, and generating personalized treatment recommendations (He et al., 2023; Y. Wang et al., 2023; Yu et al., 2023). Their potential impact is substantial, with estimates suggesting that LLMs could optimize 5%–10% of U.S. healthcare spending ($200–360 billion annually) by improving efficiency in clinical documentation and healthcare system operations (Bhasker et al., 2023; Sahni et al., 2023). The medical community has responded enthusiastically, as reflected by the rapid growth in related research, with more than 539 PubMed publications on ChatGPT applications appearing within six months of its release (Doo et al., 2023). Meanwhile, healthcare systems are increasingly adopting these technologies across a wide range of clinical and administrative tasks (Landi, 2024; Webster, 2023; Landolsi et al., 2024). In parallel with LLM-based systems, several studies have explored data-driven healthcare modeling to improve medical decision support and diagnostic efficiency. Metaheuristic-based optimization has been applied to enhance feature selection and model training in medical machine learning pipelines (Bacanin et al., 2023, 2024b). Related work also investigates optimized diagnostic models and synthetic medical data generation to address performance and data availability challenges (Zivkovic et al., 2022; Jovanovic et al., 2023; Bacanin et al., 2024a). This paradigm shift reflects AI's growing capacity to augment healthcare professionals as they navigate increasingly complex medical environments (Karabacak and Margetis, 2023).

LLMs including GPT (Brown et al., 2020), LLaMA (Alvarado et al., 2023), and Gemini (AlGhozali and Mukminatun, 2024) demonstrate impressive capabilities in general natural language processing tasks.

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However, their healthcare applications are severely limited by persistent issues with factual inaccuracies and hallucinations (Pal et al., 2023; Ji et al., 2023), which remain active areas of research (Lin and Zhang, 2023). Even specialized medical models like BioBERT (Lee et al., 2019), ClinicalBERT (Alsentzer et al., 2019), and Med-Palm (Singhal et al., 2023) struggle with contextual medical data and frequently generate biased or potentially harmful outputs, highlighting the critical need for more robust solutions in clinical settings.

Retrieval-Augmented Generation (RAG) represents an innovative approach that enhances large language models by integrating external knowledge retrieval capabilities (Kang et al., 2023). This method combines the generative power of LLMs with real-time access to authoritative medical sources including clinical databases and research literature (Kang et al., 2023). The RAG process operates through semantic similarity calculations to retrieve relevant information which then informs the model's responses (Wen et al., 2024). This architecture significantly reduces factual inaccuracies through in context learning mechanisms (Dong et al., 2024), proving particularly valuable in healthcare where accuracy and current knowledge are critical (Das et al., 2023). By incorporating up to date medical evidence, RAG enables more reliable diagnostic support and personalized treatment recommendations (C. Wang et al., 2023). The effectiveness of this approach has led to its widespread adoption across various healthcare applications (Kandpal et al., 2023; Abo El-Enen et al., 2025; R. Zhang et al., 2024).

Recent surveys have systematically investigated RAG from different perspectives. Zhao et al. (2026) examined RAG's applications in AI-generated content, proposing a unified framework for retrieval-augmentation methodologies. Hu and Lu (2024) offered a comprehensive analysis of both RAG and understanding paradigms. Fan et al. (2024) specifically investigated architectural designs for integrating RAG with LLMs. Zhao et al. (2026) developed a novel classification of RAG tasks based on query complexity levels. While these works provide valuable theoretical and technical insights, they largely overlook domain-specific applications. Within healthcare, Amugongo et al. (2025) focuses on healthcare, yet their analysis remains limited to some medical fields without addressing critical needs in important ones like mental health and medical scientific research. The authors selected 37 papers and structured their discussion primarily around three RAG categories: Naive, Advanced, and Modular. They also explored evaluation, ethical considerations, and data-related aspects in a comprehensive way. While six medical domains were extracted from the selected papers, namely Biomedical, Clinical, Gastrointestinal, Orthodontic, Pediatric diseases, and COVID-19, this classification was only used when discussing datasets. In addition, the quality of the selected papers was not considered during the selection process. As a result, the conclusions are mainly technical, addressing aspects such as context window size, general RAG enhancements, the use of multimodal data, and concerns about evaluation methods and data quality. More recently, Neha et al. (2025) reviewed 30 healthcare studies and covered applications such as diagnostic assistance, EHR and discharge note summarization, medical question answering, patient education, clinical trial matching, and biomedical literature synthesis. Similarly, Abo El-Enen et al. (2025) provide a broad overview of RAG applications in healthcare, including clinical documentation, named entity recognition, multimodal imaging, decision support, and population health surveillance. The survey also introduces a taxonomy of Basic, Advanced, and Modular RAG systems and analyzes architectures, models, evaluation benchmarks, and reported performance across existing studies. Although these surveys provide broad overviews of tasks and architectural designs, medical specialties are typically mentioned or discussed only briefly.

Table 1 presents a side-by-side comparison of existing Retrieval-Augmented Generation surveys according to medical field coverage. "Brief" indicates that a field is mentioned only through limited application examples or datasets, whereas "In-depth" denotes sustained domain-specific discussion with reference to multiple studies. Our analysis focuses on medical fields with a sufficient number of eligible studies to enable meaningful comparison. Although orthodontics appears in prior surveys, it was not retained as an independent field due to the limited number of studies meeting our inclusion criteria. As shown, only 50% of the surveyed reviews explicitly address healthcare applications of RAG, and those that do primarily concentrate on general medicine. Among healthcare-focused surveys, coverage extends to only 50% of the medical fields identified in our taxonomy as having sufficient study density for analysis. Moreover, while recent surveys provide valuable architectural and methodological overviews of RAG, healthcare applications are typically presented as illustrative examples rather than examined as domains requiring differentiated design principles. Even dedicated healthcare reviews tend to organize discussion around applications or system components without systematically analyzing how RAG configurations should adapt across distinct medical specialties. Consequently, there remains limited evaluation of how clinical objectives influence architectural suitability. Addressing this lack of structured field-level synthesis constitutes a central contribution of our review. The novelty of this review lies specifically in the depth of field-level treatment rather than the mere mention of medical applications. Prior surveys addressing healthcare typically reference medical fields only through brief illustrative examples, whereas the present review provides sustained, domain-specific synthesis supported by multiple studies per field.

To the best of our knowledge, there is no existing review that clearly focuses on RAG applications classified by medical fields while analyzing their distinct needs and the potential benefits for patients and practitioners. For practitioners specifically, this benefit takes the form of domain-level orientation, awareness of which RAG architectures have been applied in their clinical area and how well they fit specific tasks, rather than deployment or implementation guidance. Moreover, we have targeted only high-ranked papers majored by their impact factor, quartiles, or conference rankings. Accordingly, our discussions and conclusions are synthetized by medical areas rather than general technical methods, datasets, or metrics. Our objective is to provide a comprehensive, current, and domain-oriented review of RAG that highlights its contributions by medical fields. This review may serve as a valuable reference for future work applying RAG in specific medical contexts. Our analysis reveals that despite the availability of sophisticated enhancement techniques, several healthcare domains continue to rely predominantly on Basic RAG architectures, underexploring advanced pre-retrieval and post-retrieval refinements that are particularly critical for complex medical tasks requiring high accuracy and structured reasoning. Moreover, several major medical specialties remain largely unexplored by RAG research, including neurology and ophthalmology. This gap between available techniques and actual deployment represents a key finding of our review. This survey provides the following contributions:

- We provide a domain-specific systematic review of RAG applications across 10 medical fields, analyzing 57 high-quality studies.
- We introduce a structured comparative framework that categorizes systems into Basic, Advanced, and Modular architectures, a taxonomy adopted from established prior work, and analyzes their pre-retrieval and post-retrieval enhancement strategies, applying it consistently across fields to support systematic cross-domain comparison.
- We demonstrate through cross-domain synthesis that optimal RAG architecture is domain-specific and task-dependent rather than universally determined by retrieval complexity.
- We identify an over-reliance on Basic RAG configurations and analyze how limited use of retrieval enhancement techniques constrains robustness and clinical reliability.
- We highlight critical research gaps, including uneven field adoption, limited exploration in major specialties, and the need for clinically meaningful evaluation standards.

Table 1
Comparative analysis of RAG surveys by medical field coverage.

Medical field	Zhao et al. (2026)	Hu and Lu (2024)	Fan et al. (2024)	Amugongo et al. (2025)	Neha et al. (2025)	Abo El-Enen et al. (2025)	Ours
General medicine	–	–	–	Brief	In-depth	Brief	In-depth
Biomedical	–	–	–	Brief	Brief	Brief	In-depth
Radiology	–	–	–	Brief	Brief	Brief	In-depth
Mental health	–	–	–	–	–	–	In-depth
Gastroenterology	–	–	–	Brief	–	–	In-depth
Immunology	–	–	–	Brief	–	–	In-depth
Nutrition	–	–	–	–	–	–	In-depth
Pharmacology	–	–	–	–	–	–	In-depth
Research	–	–	–	–	Brief	Brief	In-depth
Education	–	–	–	–	Brief	Brief	In-depth

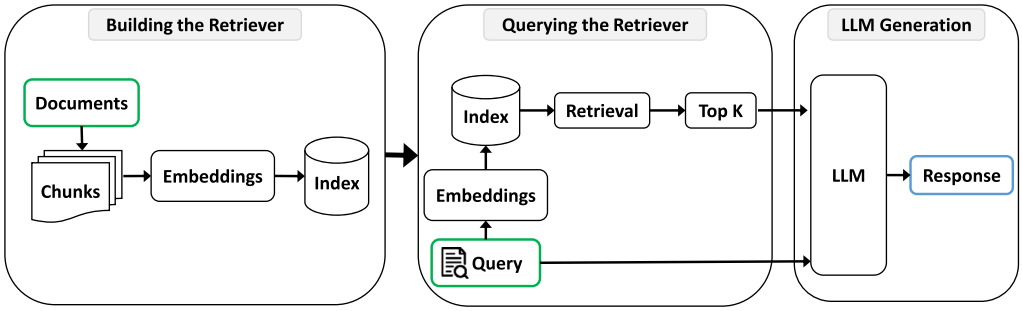


Fig. 1. A basic RAG pipeline.

The remainder of this paper is organized as follows. Section 2 defines preliminary concepts, introduces the architecture and mechanisms of RAG and discusses key enhancement techniques. Section 3 outlines the methodology for collecting and selecting survey papers. Section 4 discusses RAG methods classified by their medical application fields. The final Section 5 highlights potential research directions for future work.

2. Preliminary

2.1. Basic RAG

Retrieval-Augmented Generation (RAG) is a hybrid framework that combines retrieval-based and generative models to improve response accuracy and factual grounding in natural language processing tasks. By retrieving external knowledge and integrating it during generation, RAG mitigates hallucination and the knowledge limitations of standalone language models while avoiding the rigidity of purely retrieval-based systems.

As illustrated in Fig. 1, a typical RAG pipeline involves three main stages:

1. **Retriever Construction:** Documents are divided into smaller chunks using strategies such as fixed-length, semantic, or content-based chunking. These segments are then encoded into vector embeddings that capture semantic meaning (Abo El-Enen et al., 2025). The embeddings are stored in vector databases such as Milvus, FAISS, or LlamaIndex to enable efficient similarity-based retrieval.
2. **Query Retrieval:** The input query is encoded using the same embedding model and compared with stored document embeddings through Approximate Nearest Neighbor (ANN) search. The system retrieves the top-k most relevant passages according to similarity measures such as cosine similarity, optionally followed by reranking.
3. **LLM Generation:** The retrieved context is incorporated into the language model to guide generation. Common fusion mechanisms include query-based fusion, logits-based fusion, and latent fusion.

Despite its effectiveness, basic RAG presents several limitations:

Retrieval challenge. Retrieval based primarily on vector similarity may return redundant or partially relevant passages due to limited semantic reasoning.

Generation challenge. Basic RAG generally performs a single retrieval step, which restricts its ability to handle complex queries requiring iterative reasoning or context refinement.

Ambiguity handling. Ambiguous queries may lead to incomplete or misaligned retrieval results, which directly affects generation quality.

Over-reliance on retrieved content. The language model may rely excessively on retrieved passages rather than synthesizing new insights.

Basic RAG represents the simplest implementation of retrieval-augmented generation (Abo El-Enen et al., 2025). These systems typically lack mechanisms such as query refinement, iterative retrieval, or feedback loops, limiting their contextual depth and adaptability. To overcome these limitations, more advanced architectures introduce additional pre- and post-retrieval mechanisms.

Advanced RAG improves both retrieval and generation quality by integrating enhancement techniques before and after the retrieval step, as illustrated in Fig. 2.

2.2. Pre-retrieval

In basic RAG systems, user queries may be ambiguous or poorly aligned with indexed content. Pre-retrieval techniques aim to improve retrieval effectiveness by optimizing document indexing and query formulation (Abo El-Enen et al., 2025).

2.2.1. Indexing optimization

Several indexing strategies enhance document representation, including chunk optimization, Reverse Hypothetical Document Embedding (HyDE), hierarchical indexes, and knowledge graphs.

Chunk optimization. Chunking divides documents into manageable units for retrieval. Larger chunks capture richer semantics but may introduce noise, whereas smaller chunks reduce noise but may lack contextual completeness. Techniques such as recursive chunking and sliding windows help preserve semantic continuity. The Small2Big method retrieves sentences while incorporating surrounding context to improve semantic coverage (Monigatti, 2024).

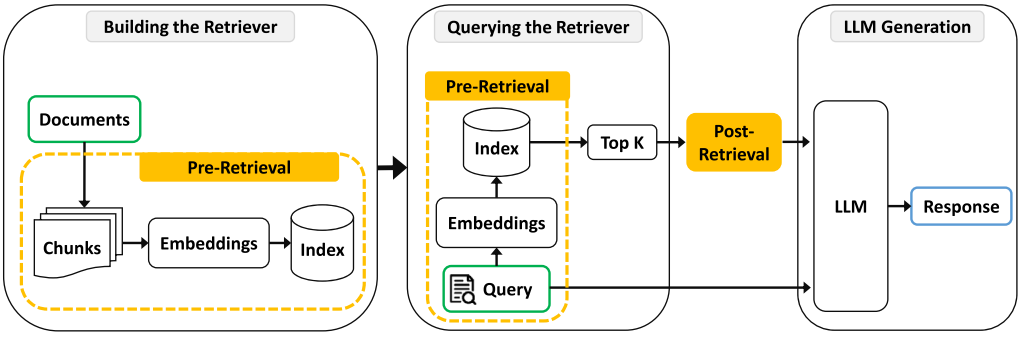


Fig. 2. Advanced RAG pipeline.

Reverse hypothetical document embedding. Reverse Hypothetical Document Embedding (HyDE) generates hypothetical answers using an LLM and embeds them to guide retrieval (Gao et al., 2023). The similarity between these generated representations and the original query is then evaluated to improve document matching (Bae, 2025).

Hierarchical indexes and knowledge graphs. Hierarchical indexing organizes documents into parent–child structures that store summaries or metadata for efficient navigation and retrieval. knowledge graphs structure document collections by modeling entities and relationships across texts (Wang et al., 2024). For example, the knowledge graph prompting framework proposed in Wang et al. (2024) constructs graph-based indexes where nodes represent textual units and edges capture semantic relations, enabling improved multi-document reasoning.

2.2.2. Query optimization

Query optimization techniques improve the alignment between user queries and document representations. Query rewriting reformulates the query to better match indexed content (X. Ma et al., 2023). Query expansion introduces related terms to broaden retrieval coverage (Monigatti, 2024). Query embedding adapts query representations using techniques such as domain-specific encoders or contrastive learning (Zhou et al., 2023). Query routers detect the query intent and select the most appropriate retrieval strategy. Metadata optimization restricts retrieval using attributes such as domain, language, time range, or document type.

2.3. Post-retrieval

Retrieved passages may include noisy or redundant information that can degrade generation quality. Post-retrieval techniques therefore refine and filter retrieved context before it is passed to the language model.

2.3.1. Prompt engineering

Chain-of-Thought (CoT) prompting has been integrated into RAG pipelines to enhance reasoning and factual consistency. By guiding the model through intermediate reasoning steps, CoT improves interpretability and generation accuracy (Wei et al., 2022). Subsequent work such as Yao et al. (2023) extends this approach by retrieving evidence for intermediate reasoning steps.

2.3.2. Reranking

Reranking re-evaluates initially retrieved passages using more precise scoring mechanisms to reorder them according to semantic relevance. Traditional approaches rely on ranking metrics such as relevance scores or Mean Reciprocal Rank (MRR) (Zhuang et al., 2023), while neural reranking methods employ transformer-based architectures such as BERT to capture contextual similarity (Abo El-Enen et al., 2025). Specialized rerankers including Cohere Rerank, bge-reranker-large, and MonoT5 further improve ranking quality in dense retrieval settings (van den Elsen et al., 2025). In addition, large language models such as GPT-3 or GPT-4 can perform reasoning-based reranking through prompt-driven evaluation.

2.3.3. Context selection/compression

Context compression reduces the input length by filtering redundant or low-relevance content before generation. Methods such as LLMLingua remove low-importance tokens using small language models while preserving semantic meaning (Anderson et al., 2022), and the Filter-Reranker framework combines lightweight filtering with LLM-based reranking to improve retrieval relevance (Y. Ma et al., 2023). Despite these enhancements, Advanced RAG architectures remain limited by tightly coupled components. Modular RAG addresses this issue by decoupling retrieval, reranking, and generation into independently configurable modules, enabling flexible optimization and advanced capabilities such as multi-hop reasoning and fact verification (Wang et al., 2025; Abo El-Enen et al., 2025).

Subsequently, we outline the main metrics used to evaluate RAG performance.

2.4. Evaluation metrics

The performance of RAG-based systems in healthcare requires multidimensional evaluation covering retrieval effectiveness, generation quality, factual consistency, and clinical validity. As noted in recent healthcare surveys (Neha et al., 2025; Abo El-Enen et al., 2025), traditional metrics such as textual similarity or classification accuracy are insufficient in medical contexts, where factual correctness and patient safety are critical. Evaluation should therefore assess whether relevant clinical evidence is retrieved, whether the generated response faithfully reflects this evidence, whether outputs are clinically valid and safe, and whether the system operates reliably under practical constraints such as latency and traceability.

2.4.1. Retrieval metrics

Retrieval metrics assess the system's ability to identify clinically relevant documents given a user query. Beyond classical classification-style metrics, ranking quality is particularly important in RAG systems because generation depends directly on the top-ranked retrieved evidence.

Accuracy represents the ratio of correctly predicted instances to the total number of instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

While commonly reported, accuracy can be misleading in imbalanced medical datasets, where high true-negative counts may inflate performance.

Precision (Park et al., 2024) measures the proportion of relevant documents among retrieved documents:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

High precision is critical in healthcare settings where presenting irrelevant or misleading documents may introduce clinical risk.

Recall (Salemi and Zamani, 2024) assesses the system's ability to retrieve all relevant documents:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

High recall is particularly important in safety-critical tasks, such as adverse event detection, where missed evidence may have serious consequences.

F1-score balances precision and recall:

$$F1Score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

However, classification-style metrics do not fully capture ranking effectiveness. Therefore, ranking-based measures are frequently employed:

- *Recall@k* evaluates whether relevant documents appear within the top-k retrieved results.
- *Mean Reciprocal Rank (MRR)* measures the inverse rank of the first relevant document, reflecting how quickly useful evidence is surfaced.
- *Normalized Discounted Cumulative Gain (nDCG@k)* accounts for both relevance and position, emphasizing highly ranked relevant documents (Neha et al., 2025).

These ranking metrics are especially important in RAG, as generation quality depends on the correctness and ordering of retrieved context.

2.4.2. Generation metrics

Generation metrics evaluate how well the language model produces responses conditioned on retrieved evidence. *BLEU* measures n-gram overlap between generated and reference texts (Papineni et al., 2002). *ROUGE-1* evaluates unigram overlap to assess content similarity (Lin, 2004).

Although widely used, BLEU and ROUGE primarily measure lexical similarity and do not guarantee factual correctness or clinical safety. A generated discharge summary may achieve high ROUGE overlap while omitting critical diagnoses or introducing incorrect medication information. Consequently, lexical overlap alone is insufficient in healthcare contexts (Neha et al., 2025). To address this limitation, domain-specific and factuality-oriented metrics have been introduced:

- *FactScore* evaluates factual alignment between generated content and reference evidence sources.
- *RadGraph-F1* measures structural overlap between generated and reference entity–relation graphs in radiology reporting, capturing clinically meaningful relational correctness.
- *MED-F1* quantifies alignment with standardized medical terminology in extraction tasks (Neha et al., 2025).
- Additionally, *Faithfulness* and entailment-based metrics assess whether generated responses are supported by retrieved documents, reducing hallucination risk.

2.4.3. Clinical-level evaluation

Retrieval and generation metrics quantify technical performance but do not establish clinical validity. Several studies therefore rely on expert-based qualitative evaluation using Likert-scale ratings of accuracy, safety, relevance, and pedagogical quality (Malik et al., 2024; Fang et al., 2025; Kurland et al., 2025). The S.C.O.R.E. framework proposed in Abo El-Enen et al. (2025) further assesses Safety, Consensus, Objectivity, Reproducibility, and Explainability through structured expert review, capturing dimensions that automated metrics cannot measure, particularly safety and decision appropriateness.

However, qualitative evaluation introduces reproducibility challenges because results depend on evaluator expertise, scoring protocols, and cohort size. Some works also apply model-based evaluation where outputs are judged by another large language model (Sung et al., 2024). Although scalable, this approach may inherit bias from the judging model and cannot replace clinician-grounded validation in high-stakes medical contexts.

Practical and safety-oriented indicators are also essential. Latency affects integration into clinical workflows, source traceability ensures transparency by linking outputs to retrieved evidence, and reporting false-positive burden and error severity remains critical in safety-sensitive domains. For instance, high sensitivity in adverse event detection may still produce disruptive false alarms when positive predictive value is low (Sun et al., 2024).

2.5. In summary

The evolution from basic to advanced and modular RAG architectures reflects a growing recognition of the limitations of early retrieval-augmented generation systems. While basic RAG models improve factual grounding by incorporating external knowledge retrieval, they remain limited in flexibility, task adaptability, and factual consistency. Advanced RAG architectures address these limitations through improved retrieval strategies, reranking mechanisms, and multi-hop retrieval. These techniques enhance the relevance of retrieved information and provide stronger contextual grounding for generation. However, they also increase system complexity, latency, and computational requirements, which may limit their suitability for real-time or large-scale deployment. Modular RAG frameworks further extend this paradigm by decoupling system components into interchangeable modules such as retrievers, generators, and routing mechanisms. This modular design enables greater flexibility, scalability, and independent optimization of components, which is particularly beneficial in multi-domain or multi-task settings. Nevertheless, coordinating multiple modules introduces additional system orchestration challenges, and overall performance remains strongly dependent on the effectiveness of the retrieval component.

Each RAG concept and component introduced here underpins the field-by-field discussion of healthcare applications throughout this survey, giving readers less familiar with RAG a shared technical vocabulary for the specialized discussions.

These architectures have been applied across multiple domains, with notable emphasis on healthcare. The following section outlines their applications in various medical fields and highlights the benefits and main challenges associated with the deployment of RAG in this context.

3. Methodology

To provide a comprehensive review of the RAG methods in different medical fields, we have retrieved recent and relevant papers from different search engines: IEEE Xplore, Science Direct, PubMed, and Google Scholar. In this work, we focus on covering important medical fields with recent high-quality papers. The literature search was conducted on 29 October 2025 across IEEE Xplore, Science Direct, and PubMed. In these databases, no explicit lower time limit was imposed; however, all relevant results retrieved were published between 2024 and 2025, reflecting the recent emergence of RAG applications in healthcare. To ensure that earlier influential works were not overlooked, an additional search was performed on Google Scholar covering the period 2020–2023, which identified a small number of relevant studies not indexed in the other databases. Preprints were excluded since sufficient peer-reviewed literature was already available to comprehensively address the review's field-level questions, and prioritizing peer-reviewed sources ensured that all included findings had passed independent quality control prior to inclusion. The query used for each search engine is presented in Table 2. We searched for keywords especially in titles and abstracts to cover papers with RAG techniques and medical applications.

This review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. Fig. 4 illustrates the PRISMA-based study selection process. The detailed process of selecting and filtering the retrieved papers is shown

Table 2
Queries used in search engines.

Search engine	Year range	Full search string
IEEE Xplore	2024–2025	(("Document title":"retrieval augmented generat*") **OR** ("Document title":"rag")) **AND** (("All Metadata":"medic*") **OR** ("All Metadata":"health*") **OR** ("All Metadata":"clinic*"))
Science Direct	2024–2025	Title: ("retrieval augmented generation" **OR** "retrieval augmented generative" **OR** "rag") **AND** ("medical" **OR** "medicine" **OR** "health" **OR** "clinical" **OR** "clinician")
PubMed	2024–2025	(Retrieval-Augmented Generation[Title/Abstract])
Google scholar	2020–2023	"retrieval augmented generation" "rag" "health" "medical" "clinical"

in Fig. 5 while mentioning the number of papers and fields in each stage. After executing all the queries, we have collected 179 papers distributed as follows: 29 papers from IEEE Xplore, 10 papers from Science Direct, 137 papers from PubMed, and 3 papers from Google Scholar. After eliminating duplicate papers, we have obtained 174 papers. We have extracted the field of each selected paper and we have obtained some non-medical fields and 20 medical fields. We have eliminated papers with non-medical fields to select only 160 papers. During the screening process, the dataset comprised 67 Q1 papers, 16 Q2 papers, 1 Q3 paper, and 1 Q4 paper according to the SCImago Journal Rank (SJR 2023), along with 11 papers published in non-ranked journals. Among the ranked journal publications, Q1 papers account for approximately 80% of the studies. To maintain a manageable corpus while emphasizing influential venues, the final selection prioritized Q1 journals. This filtering step was applied primarily to reduce the number of analyzed studies while preserving the overall structure of the literature. The predominance of Q1 venues among ranked papers, at roughly 80%, reflects the natural composition of the retrieved corpus, as queries in each database returned very few relevant results outside this range. The decision to retain only Q1 journals was a deliberate choice for corpus manageability, and its effect on reliability is further supported by the independent quality appraisal applied to all included studies. In addition, we included papers published in conferences ranked A* and B according to the Computing Research & Education (CORE) ranking database. As a result, we obtained 68 papers across 18 medical fields.

However, many of the identified fields contained too few papers to support meaningful discussion. Therefore, fields with only a small number of papers were excluded, resulting in a final selection of 57 papers across 10 medical fields: General Medicine, Biomedical, Scientific Research, Education, Mental Health, Gastroenterology, Reporting, Immunology, Nutrition, and Pharmacology. The complete list of the selected studies, including their DOI, publication year, venue, and reason for inclusion, is provided in Table 8 in Appendix A. For transparency, the full list of screened studies, including both included and excluded papers, is provided in the supplementary material. To provide clearer insight into the quality of the selected journal papers, we classified them into high- and low-quality categories based on their journal Impact Factor (IF) using SJR 2023, the same source used for quartile classification. We computed and ranked the IF scores of the journal papers according to SJR. Using a threshold of IF > 4, we identified 43 high-quality journal papers. For the dominant fields, 9/14 (64%) papers in General medicine and 8/10 (80%) papers in Biomedical were classified as high-quality.

To ensure methodological rigor beyond venue-based screening, a structured quality appraisal was conducted for all 57 included papers. Journal quartile and impact factor were used exclusively during the initial filtering stage and were not considered proxies for methodological quality. Given the heterogeneity of study designs in RAG applications for healthcare, including experimental benchmarking studies, clinical evaluations, qualitative investigations, and mixed-methods research, the appraisal framework was adapted following the principles of the Mixed Methods Appraisal Tool (MMAT, 2018 version) (Hong et al., 2018). MMAT was selected due to its suitability for systematically assessing diverse empirical study designs within health-related research domains. It was applied exclusively to empirical studies. Each study was evaluated across five methodological domains, generating a quantitative quality score (0–5). These five methodological domains are:

- **Q1. Clarity of Research Objective:** Whether the study clearly defined its research aims and the intended healthcare application of the RAG system.
- **Q2. Methodological Transparency:** Whether sufficient technical and procedural detail was provided regarding the RAG architecture, retrieval mechanisms, dataset construction, and evaluation pipeline to allow reproducibility.
- **Q3. Dataset Appropriateness and Representativeness:** Whether the evaluation dataset was adequately described and appropriate for healthcare contexts, including clarity regarding data sources, size, and population characteristics where applicable.
- **Q4. Validity and Relevance of Evaluation Metrics:** Whether performance metrics were clearly defined, statistically reported where applicable, and clinically meaningful for healthcare applications.
- **Q5. Bias and Ethical Consideration Reporting:** Whether the study addressed potential risks such as hallucinations, demographic bias, safety concerns, fairness issues, or ethical safeguards relevant to healthcare deployment.

Each criterion was scored using a three-level scale: 1 (fully satisfied), 0.5 (partially satisfied), 0 (not satisfied). This produced a total quality score ranging from 0 to 5 for each study. Based on these scores, studies were categorized into three methodological quality levels: high quality (HQ, ≥4), moderate quality (MQ, 3–3.5), and low quality (LQ, <3). Quality assessment was performed independently by two reviewers. Discrepancies were resolved through discussion until consensus was reached. The results of this quality assessment are summarized in Fig. 3 (see Table 7 in Appendix A for the detailed results). Overall, the methodological quality of the included studies is satisfactory, with 12 (24.49%) classified as high quality (HQ), 33 (67.35%) as moderate quality (MQ), and only four (8.16%) as low quality (LQ). Some limitations across studies remain related to incomplete methodological reporting or superficial discussion of certain aspects despite addressing the integration and application of RAG methods in healthcare. This appraisal serves as an independent quality-demonstration step for the finally retained corpus, not as an additional selection criterion layered on top of the venue-based filtering already performed during inclusion. With 91.84% of the included studies rated as moderate or high quality, the quality assessment showed that the retained studies were of sufficient methodological quality. Therefore, no additional quality-based filtering was applied.

Fig. 6 presents statistics on the number of selected papers per medical field. Fig. 7 shows the distribution of selected papers across years and trimesters, while Fig. 8 illustrates their distribution across publication quality levels. Table 3 reports the minimum, maximum, mean, and standard deviation (SD) of the journal Impact Factor (IF) for the papers in each field. These statistics indicate that the selected studies are largely published in high-quality journals and conferences. In particular, 92% of the papers appear in Q1 journals, with an IF exceeding 14 in the field of scientific research and an average IF above 9 in general medicine. In addition, 2 papers are published in A* conferences. Most of the selected studies were published in 2024 and 2025, accounting for more than 93% of the dataset. Earlier years are also represented, with 3 relevant papers from 2023. Overall, the

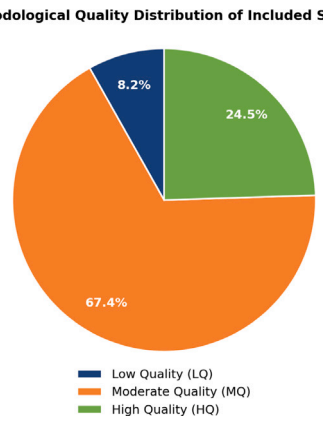


Fig. 3. Quality assessment of the included studies.

Table 3
Statistics about journal IF scores of each field.

Field	Min	Max	Mean ± SD
General medicine	1.3	13.3	9.4 ± 4.5
Biomedical	4.6	7.4	5.3 ± 1.3
Scientific research	2.0	14.6	6.2 ± 4.8
Education	1.5	4.7	3.3 ± 1.4
Mental health	2.3	2.5	2.4 ± 0.1
Gastroenterology	3.1	12.5	5.4 ± 4.0
Reporting	2.0	6.1	4.4 ± 2.2
Immunology	4.1	13.6	7.9 ± 5.0
Nutrition	4.2	5.1	4.8 ± 0.5
Pharmacology	3.3	7.3	4.6 ± 2.3

number of studies applying RAG in medicine has increased markedly in the last two years. In the first trimester of 2025, the number of publications already exceeds that of the same period in 2024 by approximately 400%, suggesting a rapidly growing research trend. This concentration reflects the actual range returned by the search queries, with no minimum publication year imposed. This confirms that RAG applications in healthcare only began emerging within this window. The review therefore captures a snapshot of an emerging field rather than a longitudinal view spanning several years of development. The dominant fields are General medicine and Biomedical, representing 26% and 19% of the papers, respectively. In the next section, we comprehensively discuss the selected works, highlighting their benefits, strengths, and potential improvements.

4. RAGS for healthcare: a concise review by medical field

First, we note that the selected studies can be categorized into 10 medical fields: General Medicine, Biomedical, Scientific Research, Education, Mental Health, Gastroenterology, Reporting, Immunology, Nutrition, and Pharmacology. In the following sections, we discuss each category in detail.

4.1. General medicine

In general medicine, the application of Retrieval-Augmented Generation (RAG) has shown considerable promise in enhancing the factuality, contextuality, and task alignment of language model outputs. Several works such as those of Ong et al. (2024), Zelin et al. (2024), and Zheng et al. (2025) adopted basic RAG configurations using chunk-level embeddings and vector-based retrieval (e.g., ChromaDB, FAISS) without additional pre- or post-retrieval modules. These systems were applied to surgical support, rare disease diagnosis, and medical triage, utilizing datasets such as simulated patient data, RareDis Corpus, and real-world consultation records. RAG has also been conceptualized as a key paradigm for customizing LLMs to authoritative domain knowledge, reducing off-the-shelf error rates of approximately 30% and improving accuracy by 39.7% when grounded in clinical practice guidelines (Kunze et al., 2025). These systems demonstrate significant improvements through the integration of external knowledge, achieving 87% context precision and up to 97% recall in physician–patient matching tasks. However, three fundamental constraints limit their applicability: (1) incomplete coverage in medical knowledge bases, particularly for rare conditions; (2) insufficient granularity in document collections that fails to capture nuanced clinical contexts; and (3) inherent challenges in modeling complex, multi-step diagnostic reasoning processes.

A prominent direction in medical RAG research centers on advanced graph-based architectures that integrate structured semantic mechanisms to enhance reasoning precision. Fuellen et al. (2025), Gilbert et al. (2024), and Zhang et al. (2025) adopt advanced RAG frameworks incorporating knowledge graphs, semantic filtering, and structured reasoning components, reporting performance gains of up to 70.55% accuracy on benchmark QA tasks and 82% accuracy in structured medical appraisal. These approaches improve semantic grounding, reduce hallucinations, and enhance task-specific precision. However, they often depend on domain-specific knowledge resources and may face scalability or portability challenges across heterogeneous clinical settings. Overall, their effectiveness appears particularly suited to scenarios where structured relational knowledge and formal standards can be explicitly encoded to guide retrieval and reasoning.

Several studies tackled multi-modal scenario by embedding complex data types and integrating heterogeneous retrieval signals. Su et al. (2025) and Hu et al. (2025) implemented document parsing pipelines over image-text datasets or OCR-derived customs documents, applying vector similarity alongside keyword-based retrieval techniques. Both employed advanced RAG features such as multi-modal re-ranking and context-driven generation. Results such as 49.4% zero-shot accuracy or 99% relevancy with proprietary models reveal the effectiveness of these systems, although the intensity of the resource and the architectural complexity pose deployment challenges in dynamic clinical environments.

Within general medicine, semantic enrichment strategies are implemented across different levels of RAG sophistication. Myers et al. (2025) employ an advanced RAG using dense retrieval with BGE embeddings and cosine similarity, while enhancing representation through UMLS-based concept mapping at pre-retrieval level, achieving up to 47.5% mean average precision. Although retrieval remains architecturally basic, semantic preprocessing significantly influences performance, with sensitivity to query formulation and dataset variability reported as limitations. In contrast, Yang et al. (2024) adopt a Modular RAG architecture integrating dense retrieval, semantic tagging, context analysis, and a multi-source fusion diagnostic model, achieving 63.87% top-5 diagnostic recall and outperforming prior approaches by 5.47%. Compared to Basic RAG configurations, the modular design enables structured coordination of retrieval and reasoning components but requires domain-specific resources and manual tool development. Overall, semantic enrichment improves retrieval robustness across architectures, yet increased modular complexity enhances task-specific precision at the cost of scalability and generalizability.

Post-retrieval prompt augmentation strategies similarly vary in architectural sophistication. Glicksberg et al. (2024) implement an advanced RAG framework that combines dense retrieval with probabilistic signal injection from an ensemble machine learning model into GPT-4 prompts, achieving up to 83.1% accuracy. This design enhances reasoning through structured contextual grounding and explicit stepwise inference. Similarly, Yazaki et al. (2024) employ an advanced RAG pipeline for emergency triage support. Their framework integrates context summarization, multi-vector retrieval over Emergency Severity Index (ESI) guideline knowledge, and Chain-of-Thought reasoning within GPT-3.5 to structure triage decision-making. The system

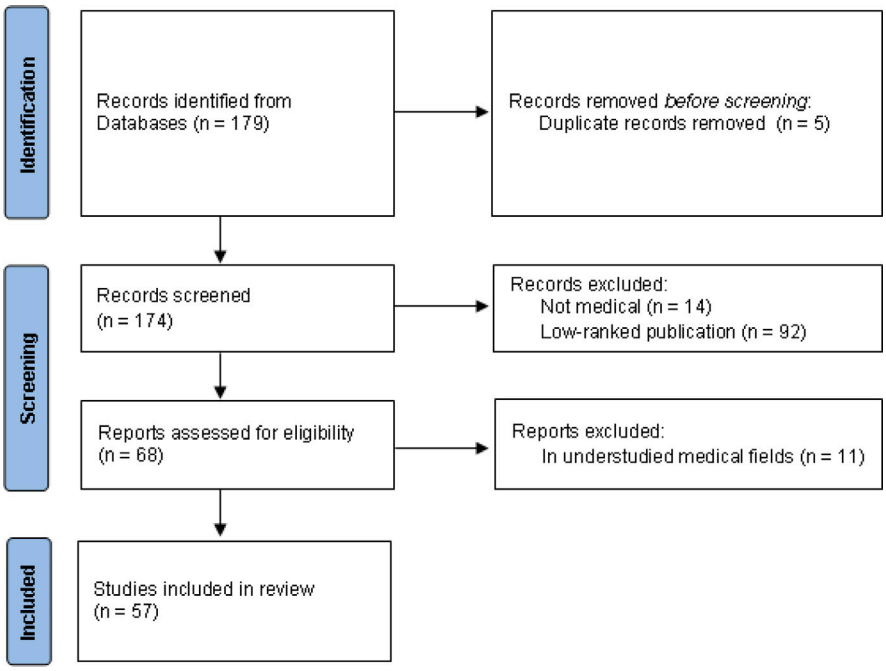


Fig. 4. PRISMA-style flow diagram of study selection process.

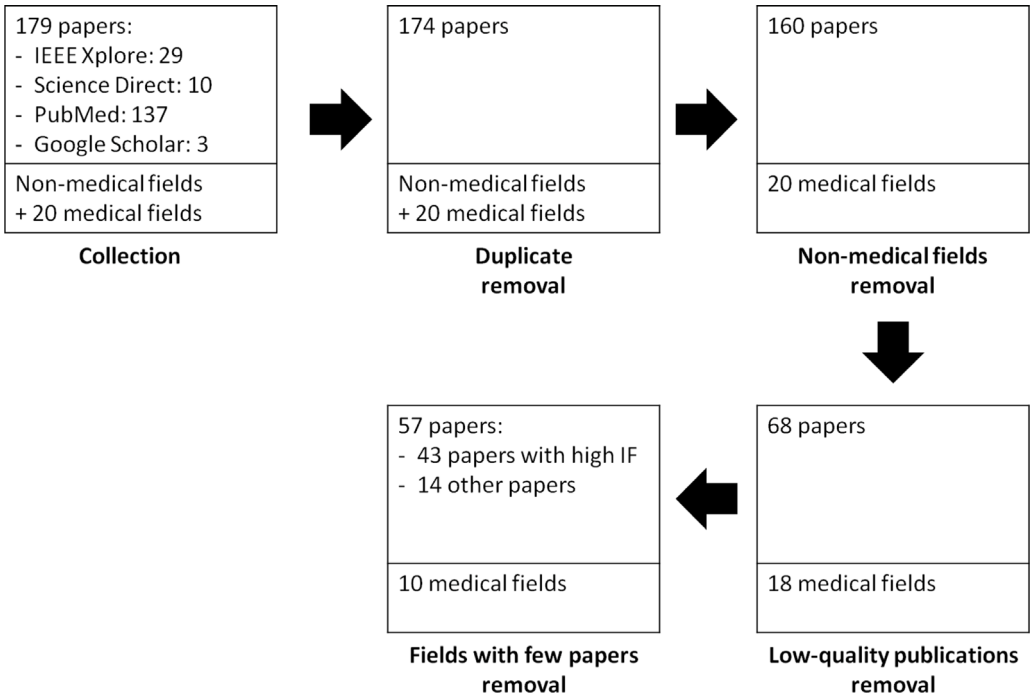


Fig. 5. The process of filtering and selecting the papers.

achieves a correct triage rate of 70%, improving triage consistency and clinical decision support. However, its evaluation relies on structured simulated scenarios without independent adjudication. By comparison, Nahass et al. (2024) rely on a basic RAG configuration using dense retrieval followed by prompt template modification. Their results show no significant improvement, with accuracy remaining at 54.3% in examination tasks. This contrast suggests that while basic RAG with prompt reformulation provides contextual grounding, more advanced RAG systems that incorporate structured retrieval strategies and explicit reasoning mechanisms can yield more substantial performance gains. Overall, the effectiveness of post-retrieval enhancement depends not only on prompt design but also on the degree of architectural integration beyond standard dense retrieval.

In summary, RAG in general medicine shows strong potential to improve clinical decision support and information retrieval. However, consistent challenges include the need for highly curated and comprehensive medical corpora, careful consideration of retrieval granularity, and robust evaluation metrics that go beyond simple accuracy to assess clinical utility and safety. Diagnostic tools in general medicine prioritize precision and verifiability, often requiring complex verification workflows.

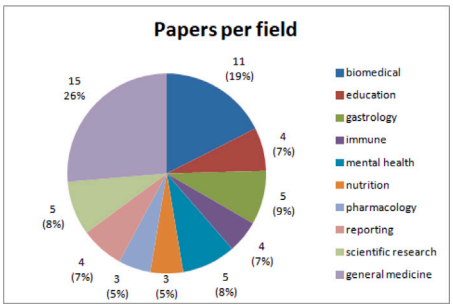


Fig. 6. Distribution of the selected papers across fields.

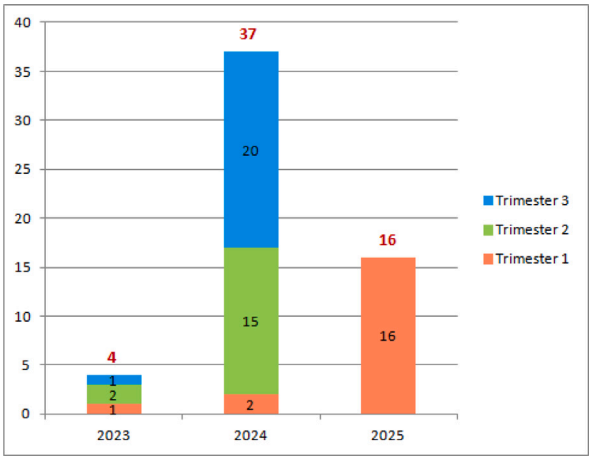


Fig. 7. Distribution of the selected papers across years.

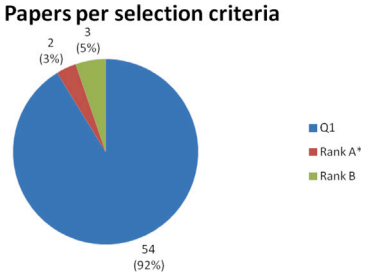


Fig. 8. Distribution of the selected papers across publication quality level.

4.2. Biomedicine

Recent advances in retrieval-augmented generation for biomedical applications highlight the performance gap between advanced RAG and modular RAG architectures, extending beyond standard retrieval methods. knowledge graph–enhanced systems have demonstrated particular promise. For instance, Matsumoto et al. (2024) and Matsumoto et al. (2025) both utilize the AlzKB dataset for question answering in combination with Weaviate embeddings. Matsumoto et al. (2024) implement an advanced RAG incorporating knowledge graph–to–text transformation at the pre-retrieval stage and Graph-of-Thoughts synthesis post-retrieval, achieving 70.4% accuracy in question answering. In contrast, Matsumoto et al. (2025) adopt a modular RAG design integrating strategy generation and Python code execution within a coordinated reasoning pipeline, attaining 93.4% accuracy. While both approaches employ dynamic Graph-of-Thoughts mechanisms to strengthen reasoning depth, the advanced RAG remains dependent on structured knowledge graph availability, whereas the Modular RAG design introduces increased computational overhead and execution latency. Collectively, these findings underscore that performance gains correlate with architectural sophistication, yet also highlight the practical trade-offs that currently limit widespread clinical deployment.

Alternative knowledge graph implementations further illustrate the performance gains and practical constraints associated with structured retrieval augmentation. Di Maria et al. (2024) adopt a modular RAG architecture that dynamically constructs query-driven knowledge graph from PubMed Central and DisGeNET using OntoTagMe, achieving 87% accuracy on disease-related tasks. While this modular coordination enhances reasoning transparency and structured inference, it requires manual validation and introduces additional workflow complexity. In contrast, Soman et al. (2024) implement an advanced RAG integrating the SPOKE knowledge graph with PubMedBERT embeddings to optimize token efficiency, reaching 95% accuracy on true/false biomedical questions. Although computationally more streamlined than fully modular orchestration, this approach remains narrowly disease-focused and dependent on domain-specific graph coverage. Collectively, these studies demonstrate that knowledge graph–enhanced RAG architectures consistently outperform standard dense retrieval. However, their benefits are coupled with domain dependency, scalability limitations, and integration overhead that may constrain broader clinical deployment.

Specialized retrieval approaches have emerged alongside knowledge graph–based systems, reflecting increasing architectural diversity. Jeong et al. (2024) introduce a modular RAG framework employing MedCPT embeddings with reflective token reranking, achieving 46.5% accuracy on MedQA while incurring notable computational overhead due to multi-stage reasoning and reranking. Similarly, Kim and Yoon (2024) implement a modular RAG configuration that integrates BM25 lexical search with neural retrieval using RoBERTa embeddings for chemical–protein interaction extraction, attaining 91.23% ROUGE-1 but requiring curated training data to maintain performance. In contrast, Cinquin (2024) adopt an advanced RAG architecture leveraging middleware-based query correction and API-driven genomic database retrieval with post-retrieval snippet synthesis, improving factual consistency without full modular orchestration. Collectively, these approaches demonstrate that moving beyond standard retrieval enhances task-specific precision, yet increased modular coordination and external system integration introduce data dependency, computational cost, and deployment complexity that may limit real-time scalability.

A large-scale meta-analysis by Liu et al. (2025) evaluated 335 biomedical studies and reported a pooled odds ratio of 1.35, indicating overall performance improvements associated with RAG-based approaches, while also noting substantial variability in evaluation metrics and study design. Complementarily, Franklin et al. (2024) examined bias mitigation through semantic ontologies, emphasizing that semantic augmentation may reduce certain biases but does not eliminate the need for human oversight. Together, these studies extend the discussion beyond individual implementations, highlighting recurring challenges in biomedical RAG research, including heterogeneous evaluation practices, domain dependency, computational constraints, and issues related to clinical translation.

In summary, biomedical RAG systems excel in leveraging structured knowledge but face hurdles in dynamic real-world scenarios due to their reliance on specific knowledge graphs and high computational demands. The field shows a clear progression from basic to advanced and modular architectures, with systems like Self-BioRAG and ES-CARGOT demonstrating sophisticated reasoning capabilities through modular component design. These systems require more adaptable and computationally efficient architectures, alongside robust validation methods, to bridge the gap between impressive research results and practical clinical deployment. Biomedical RAG systems must ensure that validation protocols are adaptable to diverse medical environments and real-time decision-making, where computational efficiency and accuracy are key elements.

4.3. Medical scientific research

In medical scientific research, RAG is primarily used for literature-centered tasks including domain-specific question answering, automated summarization, recommendation, and experimental result interrogation. This field shows a stronger presence of advanced and modular RAG compared to some other medical domains. A clear example of basic RAG is Luu and Buehler (2023) who use a corpus of 1000+ peer-reviewed papers on biological and bio-inspired materials with paragraph-level BAAI/BGE embeddings and Chroma vector retrieval, followed by generation with Orca-2-13b. No pre-retrieval or post-retrieval enhancement is reported, and the study reports a score of 90%. The main benefit is traceability for recall and hypothesis generation within the target corpus, whereas the key limitation is narrow domain coverage and minor factual slippage when retrieval alone is insufficient for verification.

Advanced RAG is more common in literature summarization and recommendation, where authors explicitly add preprocessing, ranking, and reasoning controls. Kurland et al. (2025) analyze Web of Science papers on common spinal pathologies to automatically summarize and cite salient details from scientific articles. They use OpenAI ada-embed and MedCPT chunk embeddings with embedding-based retrieval. They apply text extraction and clean-up pre-retrieval and chain-of-thought prompting post-retrieval with GPT-4, reporting 97.5% factual and citation accuracy under manual review. Li et al. (2024) present an advanced RAG system for biomedical literature recommendation and summarization using PubMed API retrieval with SentenceTransformer embeddings. GPT-4 supports keyword extraction pre-retrieval, while a multivariable ranking algorithm guides post-retrieval selection before structured summarization with GPT-4 Turbo. Expert evaluation on a 1–5 scale reports strong performance across tasks, including relevance 4.40, accuracy 4.40, and reference integration 4.60. The system reports no fabricated papers and high metadata accuracy, although it remains limited by PubMed keyword-based retrieval and token constraints. Wysocka et al. (2024) evaluated GPT-4 using Wikidata and PubMed abstracts and reported 70% factual answers for chemical entity recognition and 43.6% for chemical–fungi relations. Based on these limitations, they propose an advanced RAG extension that integrates robust RAG with high-recall retrieval in coordination with symbolic reasoning to improve factual grounding and mitigate entity bias. Overall, advanced RAG in this field improves citation grounding, but performance remains closely tied to GPT-4-class models and curated indexing resources. Overall, citation fidelity improvements reflect structured retrieval–reasoning coordination, and notably, most advanced systems in this field are implemented using GPT-4 models.

Modular RAG appears when research workflows require integrating heterogeneous inputs beyond static literature corpora. Craig and Drăghici (2024) introduce a modular RAG system designed for question answering over user experimental data. The system integrates PubMed-derived knowledge bases with user-provided experimental documents and quantitative results. Paragraph-level text-embedding-ada-002 embeddings are indexed in Pinecone, with sub-question generation applied pre-retrieval and quality assessment post-retrieval. GPT-4o is used to answer questions about specific experiment. The architecture separates domain retrieval, experimental context construction, and function-based quantitative reasoning via a user-defined REST API. The primary advantage is paragraph-level traceability with strict grounding in retrieved evidence, reducing hallucination risk while enabling integration of experimental and quantitative data. However, the need for custom REST API configuration and serial handling of complex queries constrains nonlinear reasoning and broader scalability.

Overall, RAG systems in medical scientific research demonstrate considerable promise for enhancing efficiency and accuracy in information processing. The field spans basic implementations through advanced to modular architectures. However, persistent challenges include ensuring the factual accuracy of LLM outputs, mitigating biases inherent in training data and retrieval mechanisms, and addressing the dependency on proprietary LLMs. These tools require robust validation protocols to ensure their reliability and ethical deployment in research workflows. The ability to verify and cross-check information in medical research is critical. Ensuring that the RAG system outputs remain factual and unbiased is essential for research synthesis.

4.4. Medical education

In medical education, RAG is mainly applied to examination assessment, board preparation, anatomy tutoring, and simulated teaching evaluation. Compared with several clinical domains, this field shows a stronger presence of advanced RAG, while basic implementations remain limited. Chang and Lin (2024) apply a basic RAG to bilingual emergency medicine board exams using PubMedBERT embeddings over 2000-token segments with token overlap. It is based on an ensemble reviewer retrieval with BM25 and PubMedBERT models which is followed by GPT-4 for the generation. The system achieves 69.81% accuracy. Although it improves bilingual performance and reduces response randomness, improvements remain moderate, reflecting limited enhancement beyond retrieval and generation.

Advanced RAG in medical education is predominantly applied to structured assessment tasks, where controlled preprocessing and guided reasoning are emphasized over retrieval scale. In simulated teaching evaluation, Fang et al. (2025) combine speech-to-text processing with bce-embedding-base_v1 segment embeddings indexed in Milvus, apply dynamic phrase repetition and character filtering before retrieval, and perform re-ranking with bce-remaker-base_v1 after retrieval, with local 7B models achieving expert mean scores of 4.13 for content accuracy and 4.50 for logic and organization on a 1–5 scale. In anatomy tutoring, Arun et al. (2024) use document-level embeddings and vector similarity search over curated resources including National Library of Medicine StatPearls, OpenStax Anatomy and Physiology 2E, and SEER Training, applying conversation summarization before retrieval to guide ChatGPT 3.5, yielding 77.8% accuracy and high expert ratings such as 4.78 for factual accuracy. For licensing examinations, Zhao et al. (2025) design an advanced RAG system built on ChatGPT-4 that integrates a curated nursing knowledge base. Embedding is performed by converting uploaded knowledge texts into semantic vectors to enable similarity search. Pre-retrieval enhancement consists of structured analysis of the question stem and all answer options to refine the retrieval query. Retrieval is conducted through OpenAI RAG using semantic search over the embedded corpus. Post-retrieval enhancement applies a strictly ordered prompt workflow that enforces step-by-step evaluation of each option. Finally, the GPT-4 generation stage produces the answer and explanation. Evaluated on 720 mock licensing questions, the system achieves over 90% accuracy and significantly outperforms standard GPT-4. Collectively, these studies suggest that improvements in medical education stem primarily from structured preprocessing, curated knowledge integration, and constrained reasoning rather than retrieval expansion alone. However, recurring limitations include narrow datasets, closed-form assessment formats, limited multimodal capability, and reliance on GPT-4–class models, leaving broader curricular scalability largely unresolved.

In summary, RAG applications in medical education demonstrate significant potential to improve learning effectiveness and assessment accuracy. The field showcases both advanced and basic RAG implementations. Although various LLMs and retrieval techniques have shown promising results, consistent challenges include adapting to multimodal learning content, handling diverse language contexts, and ensuring the scalability of customized knowledge bases. More comprehensive evaluation frameworks are needed to capture the pedagogical impact and explore broader applicability across different medical disciplines and educational formats. Educational RAG applications must prioritize effectiveness metrics, ensuring that the system adapts to diverse teaching styles and learning environments while maintaining accuracy.

4.5. Mental health

Chen et al. (2024) examines the transformative potential of generative AI in psychological research, focusing on behavior modeling, data analysis, and simulation of social interaction. Rapid advancement of tools such as ChatGPT offers unprecedented opportunities to handle large datasets and increase research efficiency. However, significant theoretical and ethical challenges emerge regarding political biases, geographic biases, and the need for coherent theoretical frameworks. Proposed solutions include Retrieval-Augmented Generation approaches and human-in-the-loop systems to maintain scientific rigor while leveraging AI capabilities.

Basic RAG appears in diagnostic accuracy assessment. Gargari et al. (2024) evaluate psychiatric case reasoning over DSM-5 Clinical Cases. Embedding is conducted with BGE at chunk level. Retrieval supplies relevant passages without explicit pre- or post-retrieval enhancement. LLM generation using GPT-3.5 Turbo and Nemotron-3-8B achieves 13/20 and 7/20 correct diagnoses respectively. The approach demonstrates feasibility for structured psychiatric interpretation but is constrained by small sample size and case variability.

Advanced RAG is observed in instruction dataset construction rather than direct inference. Sung et al. (2024) generate psychiatry Q&A data from the Desk Reference to DSM-5. Embedding is performed with Nomic Embed at chunk level. Pre-retrieval enhancement converts source material into structured JSON to standardize queries. Retrieval uses FAISS similarity search. LLM generation with Mistral-7B produces instruction–response pairs. The generation is evaluated by GPT-4 with a score of 726/800 after summing scores between 1–10 of 80 questions. The strength lies in dynamic data synthesis to address domain scarcity, though applicability remains limited to psychiatry.

Modular RAG appears in structured clinical extraction and affect-aware dialogue. Lopez et al. (2025) design an entity-aware RAG system over the CheXpert and Stanford MOUD EHR datasets, including records of patients treated for opioid use disorder. Pre-retrieval enhancement performs named entity recognition followed by ontology- and LLM-based entity augmentation. Embedding uses Bio + Clinical BERT to encode the extracted entities. Retrieval is executed through a regular expression–based tool, Target Matcher, rather than vector similarity search. LLM generation for clinical information extraction is performed using GPT-4, which achieves 97% F1-score, outperforming Mistral, while reducing token usage by more than 70% and lowering inference time. The modular separation between entity processing and generation enables plug-in LLM substitution, but the system is restricted to 18 predefined clinical variables. Similarly, Nayinzira and Adda (2024) implement a modular RAG model for counseling dialogue. Embedding is conducted at chunk level, indexed with FAISS. Retrieval relies on multi-query RAG. Post-retrieval enhancement applies a sentiment-analysis ranker before LLM generation with Mistral, yielding correctness of 82% and relevancy of 80%. While the ranking layer improves affect alignment, it increases latency and token consumption. Collectively, these modular systems demonstrate that in mental health applications, explicit separation of entity processing, retrieval, ranking, and generation enhances clinical variable extraction and emotion-aware dialogue, but scalability remains constrained by predefined variable scope and computational overhead.

Overall, RAG systems show significant promise in improving the extraction of clinical information and support in mental health. The field demonstrates a progression from basic through advanced to modular architectures. Key challenges include ensuring the reliability of diagnostic assistance, mitigating biases in language models, and developing robust sentiment analysis capabilities for nuanced emotional understanding. More comprehensive evaluation frameworks are needed to address ethical and effective deployment in this sensitive field. Therapeutic applications in mental health require specific safety measures and sensitivity to the emotional and psychological needs of patients. RAG systems must prioritize a therapeutic alliance to enhance trust and patient outcomes.

4.6. Gastroenterology

In gastroenterology, RAG implementations are approximately evenly divided between basic and advanced configurations. Basic configurations rely primarily on retrieval over curated clinical guidelines without pre- or post-retrieval control. Malik et al. (2024) use GPT-4 with retrieval through OpenAI's browser interface to answer acute liver failure queries grounded in American College of Gastroenterology guidelines. The system achieves Likert scores of 4.01 for relevance and 3.67 for accuracy. It improves guideline-backed response relevance, but potential factual inconsistencies remain and physician oversight is still required. Similarly, Giuffrè et al. (2024) apply chunk-level text-ada-002 embeddings with similarity-based vector retrieval over ESGE, EASL, and UEG guidelines, reaching 90% accuracy. While this strengthens guideline alignment, performance is constrained by context window limits and limited handling of structured tables and complex formatting. Collectively, these basic systems demonstrate that guideline-grounded retrieval can enhance clinical consistency, yet their effectiveness remains dependent on document structure and lacks deeper reasoning or refinement mechanisms to ensure robust generalization.

Advanced implementations introduce embedding optimization, structured input control, or post-retrieval filtering. Fu et al. (2024) combine internal departmental data and 17 gastroenterology guidelines for digestive endoscopic nursing Q&A. Embedding uses text-embedding-ada-002 at chunk level, retrieval relies on similarity search, and post-retrieval enhancement applies LLM-based filtering of irrelevant chunks before generating the answer with GPT 3.5. It achieves 70% accuracy and 94.5% level of detail indicating strong response completeness and reduced contextual noise. However, its effectiveness depends on a document set and external API constraints. Zhou et al. (2024) optimize retrieval through pre-retrieval fine-tuning of the gte-base-zh embedding model over 512-character chunks derived from the EGD Database containing 25 guidelines and 40 papers. Retrieval uses similarity search, followed by LLM generation with GPT 3.5, reporting 95% context recall and 93.73% faithfulness. This embedding adaptation strengthens domain-specific semantic alignment and contextual precision, though its evaluation remains geographically focused and limited in scale. Kresevic et al. (2024) restructure EASL HCV guidelines as a pre-retrieval enhancement to improve interpretability and generate responses with GPT-4 Turbo with a custom prompt engineering, achieving 99% accuracy. However, it is confined to HCV and does not report detailed chunking. Collectively, advanced systems in gastroenterology demonstrate that performance gains arise from embedding adaptation, structured guideline reformatting, and post-retrieval filtering rather than retrieval scale alone, though applicability remains confined to specific guideline sets and curated corpora.

In summary, RAG applications in gastroenterology show strong potential for enhancing Q&A systems and clinical guideline interpretation. The field includes both advanced and basic RAG implementations. However, common limitations include dependencies on specific document sets, geographic biases in specialized models, and challenges in processing complex data formats like tables and figures. These systems require more flexible data integration methods and improved multimodal processing capabilities. RAG applications in gastroenterology must ensure that clinical decision-making remains precise and based on the most relevant and diverse datasets.

4.7. Reporting

At a broader level, Doo et al. (2023) evaluate the clinical translation potential of generative models such as ChatGPT in radiology applications. Five major use cases are identified across different complexity levels, from patient communication to population health analysis. Although they show promise in improving report accessibility and workflow efficiency, these models carry substantial risks of hallucination

and require careful validation. The analysis underscores the need for radiologists to maintain oversight and develop robust safety protocols prior to clinical implementation. Furthermore, Bhayana et al. (2024) review RAG applications in radiology board examination contexts and emphasize the absence of standardized evaluation frameworks and limited ethical analysis. The analysis focused on proprietary models such GPT-3.5/4 as the most commonly used LLMs in healthcare RAG applications. The review concluded that RAG has significant potential to revolutionize healthcare applications, but requires further research to address challenges in information integration, noise handling, and verification of source factuality.

Basic RAG appears in text-centered diagnostic support. Fink et al. (2025) augment GPT-4 Turbo for trauma radiology using sentence-level text-embedding-ada-002-v2 embeddings with cosine similarity retrieval over the RadioGraphics top ten reading list. The model first retrieves relevant reference material, generates candidate primary diagnoses, and then performs a second retrieval-guided generation step for injury classification and grading. The system achieves 100% diagnostic accuracy and 96% correct classification. Its main strength lies in transparent, source-linked reasoning. However, performance is constrained by the narrow scope of the curated reading list, the added complexity of a two-stage retrieval–generation workflow, and the absence of additional retrieval refinement or ranking mechanisms.

Advanced RAG is more evident in multimodal report generation. Bernardi and Cimitile (2024) integrate retrieval with the EALBEF embedding model on the MIMIC-CXR dataset. Pre-retrieval optimization includes organ-level semantic segmentation and triple contrastive loss to align image–text representations, followed by cosine similarity retrieval. Thus, LLaVa-Med generates the radiology report. The model achieves 43% BLEU and 49% RadGraph F1. This approach reduces hallucinations through structured multimodal alignment, yet remains dependent on segmentation quality and dataset-specific tuning.

In summary, RAG systems have significant promise in improving radiology reporting and diagnostic accuracy by mitigating hallucinations and providing transparent sourcing. The implemented systems range from basic to advanced. However, persistent challenges include ensuring the quality of image segmentation, expanding the scope of knowledge bases, and establishing standardized evaluation frameworks that address both technical performance and ethical implications. Robust safety protocols and more comprehensive integration of multimodal data are essential for reliable clinical translation. Radiology reporting focuses heavily on diagnostic concordance and ensuring the accuracy and transparency of medical data used for decision-making.

4.8. Immunology

Gong and Bang (2024) critically assesses the role of ChatGPT in providing information on inflammatory bowel disease. Current evaluation methods relying on simple question-and-answer formats prove to be inadequate for clinical applications. More sophisticated approaches combining reasoning modules with RAG show potential to enhance accuracy and reliability. The work emphasizes the need for evaluation frameworks that assess contextual understanding and practical utility beyond basic factual recall in specialized medical domains.

In immunology-oriented applications, basic RAG is applied to both biological sequence modeling and clinical event detection. Chuang et al. (2025) predict IL-6–inducing epitopes using ProtTrans, a protein language model, to generate amino acid sequence embeddings for candidate peptide epitopes from the Immune Epitope Database. Retrieval is performed through vector similarity over these protein embeddings. Post-retrieval enhancement averages embeddings of similar peptide sequences, which are then used as augmented input features for multi-window convolutional neural networks. The model achieves 85.87% balanced accuracy and 90.74% area under the curve. This represents a configuration where retrieval serves feature augmentation rather than linguistic grounding. Its advantage lies in strengthening peptide representation through similarity-based enrichment, while validation remains limited to IL-6 induction and requires broader assessment for cytokine storm prediction.

Similarly, Sun et al. (2024) apply basic RAG to detect severe immune-related adverse events in 7555 hospital admissions. Vector database retrieval supports Mistral OpenOrca generation, achieving 98.1% sensitivity and 95.7% specificity, with 100% negative predictive value but 17.1% positive predictive value. The system reduces manual chart review time from 15 min to 9.5 s and identifies missed cases in 45% of false positives. Its strength lies in high sensitivity and efficiency for inpatient detection, though precision is limited by false positives linked to historical documentation.

In summary, RAG systems in immunology demonstrate significant potential for complex predictive tasks and precise identification of clinical events. Current implementations are primarily basic RAG systems. However, critical challenges involve rigorous validation of predictive models for real-world risks such as cytokine storms, and effective mitigation of false positives in highly sensitive clinical contexts. These systems require broader applicability across diverse immunological conditions and more refined techniques for disambiguating historical data from current patient conditions to enhance clinical utility. Immunology requires careful validation to avoid risks in predictive models, especially when handling complex and sensitive data and making clinical predictions.

4.9. Nutrition

In nutrition-related applications, RAG is primarily implemented in basic configurations across structured extraction, multimodal recommendation, and clinical summarization tasks. Zhan et al. (2025) combine sentence-level embeddings using MedCPT and Contriever with BM25 retrieval over a Clinical Data Repository to extract dietary supplement relations. Then, Llama2-13B and MedAlpaca-7B perform relation extraction within a multi-task learning framework, achieving 93.74% F1-score. The system benefits from shared-task efficiency and improved relation modeling, yet lacks a domain-specialized retriever tailored specifically to dietary supplements, potentially limiting retrieval precision. Similarly, Soekamto et al. (2025) integrate vision–language modeling with RAG for personalized recipe recommendation from online recipe sources. GPT-4o performs ingredient detection with 83% precision while retrieved recipe content supports recommendation generation. The approach enables multimodal personalization by linking visual input to structured recipe knowledge, but remains dependent on internet connectivity and external data availability, affecting robustness and scalability. Alkhalaf et al. (2024) apply sentence-transformer embeddings over 600-character EHR chunks with maximal marginal relevance retrieval to support zero-shot summarization of malnutrition-related records from 40 aged-care facilities. Llama 2 13B generates summaries, achieving 99.25% accuracy for explicit information extraction. While effective for structured and clearly stated data, performance degrades for implicit clinical cues, reflecting ongoing hallucination and inference limitations in basic RAG pipelines.

In summary, RAG applications in nutrition show strong capabilities in information extraction and personalized recommendations, particularly with Llama-based architectures. All current implementations in this domain are basic RAG systems. Consistent limitations include the lack of specialized retrievers for nuanced dietary information, dependency on external factors such as internet connectivity for real-time applications, and the persistent risk of hallucinations for implicit data. More tailored retrieval mechanisms and methods to mitigate hallucination risks are necessary to enhance reliability in nutrition-focused RAG systems.

4.10. Pharmacology

In pharmacology, basic RAG configurations are primarily supporting interpretability and toxicity assessment, while modular architectures target adverse event extraction. X. Zhang et al. (2024) propose a

basic RAG to combine Gene2Vec gene-level embeddings with XGBoost and predict bispecific antibody target pairs from single-cell RNA-seq data. GPT-4 is applied through prompt engineering to generate explanatory reports. This system achieves 89.29% AUC in predicting viable target pairs. The authors highlight how RAG-enhanced explanations help pharmaceutical researchers understand target pair rationales beyond simple predictions. However, the study notably excluded trispecific antibodies, limiting its applicability to more advanced therapeutic formats. Similarly, Connor et al. (2025) assess GPT-4 for organ-specific drug toxicity classification using FDA drug-labeling documents as the reference standard across liver, kidney, and cardiac endpoints. The study compares a general conversational prompt with an expert-engineered prompt that embeds structured labeling sections such as Boxed Warnings and Adverse Reactions. GPT-4 achieves moderate performance, with per-organ accuracies ranging from 64%–75% under expert prompting and lower performance under general prompting. The structured prompting improves sensitivity and overall accuracy. The authors therefore propose basic RAG to leverage knowledge embedded in GPT-4 and improve the toxicity classification. Collectively, basic RAG in pharmacology functions mainly as a grounding and explanation layer to improve transparency and safety alignment, while predictive performance still depends heavily on underlying models and curated regulatory sources.

In contrast, Wu et al. (2025) implement a modular RAG system for adverse event extraction from the FDALabel database. Their framework combines semantic and exact section-level retrieval with Llama 3.1-70B generation, achieving 96.7% accuracy and 97.8% F1 for drug-induced liver injury classification, and 93.1% F1 for cardiotoxicity. The explicit separation between retrieval, templating, and generation improves consistency with manual curation and enables plug-in LLM substitution, although reliance on domain-specific knowledge engineering and static SQL queries constrains adaptability to novel query formulations.

In summary, while RAG systems enhance pharmacological research through domain-specific knowledge integration, they face significant constraints. The field spans basic to modular architectures. Current implementations struggle with data heterogeneity in drug documentation and inconsistent reporting formats across regulatory sources. The technology shows limited adaptability beyond structured tasks, with performance heavily dependent on source quality and retrieval methodology. Key limitations include incomplete coverage of therapeutic targets (e.g., exclusion of tri-specific antibodies), static retrieval architectures that hinder dynamic query handling, and inherent tensions between maintaining specialized precision and achieving broader applicability. These constraints underscore the need for more robust solutions to handle pharmacology's complex knowledge landscape effectively.

4.11. Discussion

4.11.1. Cross-domain architectural synthesis

The RAG applications in each medical field are summarized in the comparison Tables 4 and 6. The numerical results reported are directly cited from the source studies, as indicated in the reference column. Because the included works differ substantially in datasets, benchmarks, and evaluation metrics, no weighted averaging or statistical aggregation was conducted. Instead, each study's results are reported as originally published and discussed qualitatively across different RAG architectures. This heterogeneity highlights the need for standardized, clinically meaningful evaluation protocols for RAG in healthcare. We have categorized the pre-retrieval enhancement techniques into: Query Refinement, Knowledge Graph Integration, Document Embeddings, Semantic Tagging, Pre-retrieval Analysis, Data Structuring & Formatting, Content Cleaning, and Context Summarization. The post-retrieval enhancement techniques are categorized into: Reranking, Advanced Reasoning, Knowledge Graph Operations, Prompt Engineering, Filtering, and Content Generation.

Although 57 studies were analyzed, this review identifies three critical limitations that constrain the current development of RAG in healthcare. First, adoption across medical fields remains highly uneven, with general medicine and biomedicine accounting for nearly half of the included studies, while domains such as education, mental health, gastroenterology, reporting, immunology, nutrition, and pharmacology remain comparatively underrepresented. Second, even within fields where RAG has been implemented, most systems rely predominantly on basic architectures, whereas modular designs remain substantially underutilized despite their demonstrated advantages in structured and knowledge-intensive tasks. This over-reliance on basic architectures is particularly prevalent in nutrition and immunology applications. Third, implementation challenges persist across domains, including computational constraints affecting multi-modal and knowledge graph integration, corpus incompleteness in niche specialties dealing with rare conditions, and evaluation inconsistency where non-standard metrics hinder cross-system comparison. These fundamental limitations must be addressed before RAG can achieve widespread clinical adoption.

Diverse LLM architectures were used to enhance the RAG pipeline. In addition to generating responses, many studies leverage LLMs for retrieval enhancement, particularly during the pre-retrieval phase for query refinement or the post-retrieval phase for re-ranking, filtering, and summarizing results. Also, some studies use encoder-only LLMs in pre-retrieval phase for NER or relation extraction. For response generation, GPT-4 and GPT-3.5 remain the most widely used models due to their broad medical knowledge although they are not open-source. However, some preferences are observed for some specialties. Nutrition applications show a strong preference for LLaMA-based architectures, while mental health systems frequently utilize Mistral variants, suggesting that these models may have inherent advantages for their respective domains. For the gastroenterology field, all reviewed implementations rely on GPT variants, indicating potential unexplored opportunities for model specialization in this area. However, promising open-source LLMs such as Deepseek warrant further exploration. An important counterintuitive finding is the strong performance of smaller fine-tuned models in constrained tasks like exam question answering, where they frequently outperform larger general-purpose models. This challenges conventional assumptions about model scaling and suggests that optimal model selection requires careful consideration of task requirements rather than simply choosing the largest available model.

Building on Tables 4 and 6, as well as the normalized quantitative overview in Table 5, our analysis shows that 35.41% of the selected studies employ Basic architectures. Notably, immunology and nutrition are composed exclusively of Basic configurations. Despite clear evidence that Advanced refinements improve reasoning control and contextual precision, a substantial proportion of studies have not explored these enhancements. Modular strategies account for only 18.75% of the reviewed systems and remain underutilized, even though they demonstrate strong potential for orchestrating heterogeneous components and enabling adaptive reasoning across complex scenarios, particularly when computational overhead and latency are effectively managed. Biomedicine clearly favors Advanced and especially Modular architectures, with 0% Basic systems and 57.1% Modular configurations. This distribution reflects the need to integrate structured entities, curated resources, and coordinated multi step reasoning, frequently supported by knowledge graphs and explicit orchestration mechanisms. General medicine exhibits the greatest architectural diversity but remains predominantly Advanced at 61.53%, suggesting that when workflows require explicit validation and mitigation of unsupported conclusions, performance gains arise primarily from structured reasoning and semantic verification rather than from simply increasing dense similarity. In medical education, Advanced architectures account for 75% of systems, while Modular approaches are nearly absent. This pattern indicates that examination oriented tasks with constrained formats benefit mainly from procedural control through decomposition and stepwise verification, without requiring complex multi component integration.

Table 4
Comparison of RAG applications across medical fields.

Field	Ref.	Main idea	Objective	Dataset	Retrieval technique	Retrieval enhancement		LLM
						Pre-retrieval	Post-retrieval	
General medicine	Ong et al. (2024)	RAG-enhanced LLM for surgical decision support	Diagnostic accuracy	Simulated patient data	ChromaDB (vector store)	None	None	Llama 2/3
	Fuellen et al. (2025)	RAG with knowledge graphs for geroprotective intervention evaluation	Q&A, intervention analysis	Benchmark datasets (not named)	Not specified	Knowledge Graph Integration	None	GPT-4, Claude Opus
	Gilbert et al. (2024)	RAG with knowledge graphs to reduce LLM hallucinations	Information structuring	SNOMED CT, HPO	Knowledge graph-based retrieval	None	Knowledge Graph Operations	Custom LLM framework
	Su et al. (2025)	Hybrid RAG with multi-modal metrics for Internet of Medical Things	Diagnostic accuracy	PMC-15M (600 K image-text pairs)	Vector similarity + keyword search	None	Reranking	LLaVA-Med
	Zhang et al. (2023)	Augments small LMs with external knowledge for knowledge-intensive tasks	Q&A	MedQA-USMLE, StrategyQA	BM25	Query Refinement	Reranking	Flan-T5
	Yang et al. (2024)	A conversational AI agent for rare diseases combining RAG with multi-source fusion diagnosis.	Knowledge Q&A and medical consultations for rare disease diagnosis.	Orphanet, OMIM, GARD, Orphadata	FAISS (Dense retrieval)	Semantic Tagging	Advanced Reasoning	GPT-3.5-turbo
	Zheng et al. (2025)	Enhancing physician recommendations using RAG for web-based medical services	Diagnostic accuracy	646,383 consultation records from Xiamen University	Maximal Marginal Relevance (MMR)	None	None	GPT-4o
	Hu et al. (2025)	Combines multimodal document parsing, hybrid storage, and RAG for customs clearance	Intelligent customs clearance assistance	Chinese customs materials (tariff chapters, laws, ordinances)	Sparse (BM25) + Dense (FAISS with HNSW)	Content Cleaning, Query Refinement	None	Series of GPT, Claude, LLaMA, and Mistral.
	Myers et al. (2025)	Comparison of embedding models and pooling strategies for clinical RAG	Information retrieval	MIMIC-III, UW EHR	Cosine similarity	Semantic Tagging	None	Not specified
	Glicksberg et al. (2024)	Improving ED admission prediction with RAG-augmented GPT-4	Predictive modeling	EHR data from 7 NYC hospitals	FAISS (cosine similarity)	None	Advanced Reasoning	GPT-4
	Zhang et al. (2025)	Combines RAG with graph-based knowledge bases and deep learning for injury appraisal	Forensic injury appraisal	26,199 real-world injury appraisal cases	Graph-based knowledge base + RoBERTa-CNN	Knowledge Graph Integration	Prompt Engineering	GPT-4
	Nahass et al. (2024)	Evaluating RAG for plastic surgery exam performance	Q&A	ASPS Plastic Surgery In-Service Exam	FAISS (cosine similarity)	None	None	GPT-3.5-Turbo
	Zelin et al. (2024)	Enhancing ChatGPT for rare disease diagnosis using RAG	Diagnostic accuracy	RareDis Corpus (717 rare diseases)	FAISS (cosine similarity)	None	None	GPT 3.5
	Yazaki et al. (2024)	Integrating RAG with a triage guideline database	Emergency patient triage decision support	100 simulated triage cases (Japanese EMT exam–derived) + ESI guidelines	Multi-vector retriever	Context Summarization	Prompt Engineering	GPT-3.5

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Table 4 (*continued*).

Field	Ref.	Main idea	Objective	Dataset	Retrieval technique	Retrieval enhancement		LLM
						Pre-retrieval	Post-retrieval	
	Matsumoto et al. (2024)	Knowledge graph-enhanced RAG with dynamic Graph-of-Thoughts (GoT)	Biomedical problem-solving	AlzKB	Hybrid search	Knowledge Graph Integration	Advanced Reasoning	BioGPT
Biomedicine	Matsumoto et al. (2025)	Dynamic Graph-of-Thoughts + Cypher queries for biomedical knowledge graph integration	Clinical decision-making	AlzKB (Memgraph Knowledge Graph)	Cypher queries	Query Refinement	Advanced Reasoning	GPT-4o Mini
	Cinquin (2024)	Middleware correcting LLM queries via database API integration	Bioinformatics data analysis	NCBI, Ensembl, UniProt	REST/GraphQL/SQL	Query Refinement	Content Generation	GPT-4o
	Jeong et al. (2024)	Introduces Self-BioRAG for biomedical reasoning via retrieval, self-reflection, and domain-specific adaptation.	Medical Q&A, diagnostic accuracy	MedQA, MedMCQA, MMLU, LiveQA, MedicationQA; Biomedical corpus (PubMed, PMC, Clinical Guidelines, Medical Textbook)	MedCPT	None	Reranking	LLaMA2 (fine-tuned)
	Kim and Yoon (2024)	Combines neural search with RAG for biomedical interaction extraction	Information retrieval, Q&A	PubMed abstracts, ChemProt, DrugProt	Hybrid BM25 + neural search	Semantic Tagging	Content Generation	GPT-3.5
	Perkins et al. (2025)	Improves LLM accuracy in microbiology Q&A using RAG with clinical reference text	Q&A	Wu's Self Assessment Q&A in Clinical Laboratory Science	Built-in RAG functionality of ChatGPT by uploading files	None	None	GPT-4, Claude3
	Di Maria et al. (2024)	Builds biomedical knowledge graphs from texts and uses Graph-RAG for summarization	Information retrieval	PubMed Central, DisGeNET	OntoTagMe + SpaCy for relation extraction	Knowledge Graph Integration	None	gpt- 3.5-turbo-instruct
	Soman et al. (2024)	Integrates SPOKE knowledge graph with LLMs for token-optimized RAG	Q&A	DisGeNET, MONDO, SemMedDB	SPOKE knowledge graph REST API + embedding-based pruning	Semantic Tagging, Knowledge Graph Integration	Knowledge Graph Operations	GPT-3.5-Turbo
Scientific Research	Wysocka et al. (2024)	Framework to streamline human expert evaluation of LLMs for factual knowledge in biomedicine	Information retrieval	Wikidata Knowledge Graph, PubMed abstracts	Not specified	None	None	GPT-4
	Luu and Buehler (2023)	Fine-tuned conversational LLM for biological materials mechanics	Q&A, hypothesis generation	Corpus of 1000+ peer-reviewed papers	Chroma vector embedding database	None	None	Orca-2-13b
	Kurland et al. (2025)	Combines bibliometrics with LLMs for automated literature summarization	Information retrieval	Web of Science papers	Embedding-based retrieval	Content Cleaning	Prompt Engineering	GPT-4
	Li et al. (2024)	PubMed retrieval + multivariable ranking algorithm + GPT-4 turbo summarization	Literature recommendation and summarization	PubMed	PubMed API	Query Refinement	Reranking	GPT-4 Turbo
	Craig and Drăghici (2024)	Extensible tool for LLM interrogation of user experimental results	Q&A, data integration	PubMed papers, user experimental data	Pinecone vector database	Query Refinement	Filtering	GPT-4o

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Table 4 (*continued*).

Field	Ref.	Main idea	Objective	Dataset	Retrieval technique	Retrieval enhancement		LLM
						Pre-retrieval	Post-retrieval	
Education	Fang et al. (2025)	Evaluating simulated teaching audio for teacher trainees using RAG and local LLMs	Audio analysis of simulated teaching	Simulated teaching audio from 10 participants	Milvus vector database.	Pre-retrieval Analysis, Content Cleaning	Reranking	InternLM2-Chat-7B, Baichuan2-7B-Chat, Qwen-7b-Chat
	Arun et al. (2024)	Custom AI chatbot for anatomy education using RAG with conversation summarization	Q&A for anatomy education	National Library of Medicine (StatPearls), OpenStax Anatomy & Physiology 2E, SEER Training	Vector similarity search	Context Summarization	None	ChatGPT 3.5
	Chang and Lin (2024)	Enhancing AI performance in bilingual emergency medicine board exams using RAG	Q&A for bilingual medical exams	Taiwan's emergency medicine board exams (2015–2023)	Ensemble reviewer (BM25 + PubMedBERT)	None	None	GPT-4.0
	Zhao et al. (2025)	Leveraging prompt engineering in custom GPT for assessing Chinese Nursing Licensing Exam	Q&A for nursing exams	720 questions from three mock exams for 2024 Chinese Nursing Licensing Exam	Semantic search (OpenAI RAG)	Pre-retrieval Analysis	Advanced Reasoning	Custom GPT (based on ChatGPT-4)
Mental health	Lopez et al. (2025)	Clinical entity augmented retrieval for EHRs	Clinical information extraction	Stanford MOUD, CheXpert	Target Matcher	Semantic Tagging	None	GPT-4, Flan-T5, Med42, Llama-3, Mistral
	Sung et al. (2024)	Generates instruction datasets via RAG	Psychiatry Q&A	Desk Reference to DSM-5	FAISS	Data Structuring & Formatting	None	Mistral-7B
	Gargari et al. (2024)	Evaluating LLMs+RAG for psychiatric diagnosis	Diagnostic accuracy assessment	DSM-5 Clinical Cases	Not specified	None	None	GPT-3.5 Turbo, Nemotron-3-8B
	Nayinzira and Adda (2024)	RAG with sentiment analysis	Mental health support	Mental Health Counseling Conversations	Multi-query RAG	None	Reranking	mistral-large-latest
Gastroenterology	Fu et al. (2024)	Combines LLM with RAG for digestive endoscopic nursing Q&A	Q&A	Internal departmental data + 17 gastroenterology guidelines	Similarity search (top-10 chunks)	None	Filtering	gpt-3.5-turbo-16k
	Zhou et al. (2024)	Fine-tunes embedding model for Chinese GI disease Q&A	Q&A	EGD Database (25 guidelines + 40 papers)	Similarity search (top-3 chunks)	Document Embeddings	None	gpt-3.5-turbo
	Kresevic et al. (2024)	Reformats HCV guidelines for LLM interpretation	Diagnostic accuracy	EASL HCV guidelines	Not specified	Data Structuring & Formatting	None	GPT-4 Turbo
	Malik et al. (2024)	GPT-4 + RAG with American College of Gastroenterology guideline integration	Q&A for acute liver failure	ACG guidelines	OpenAI browser interface	None	None	ChatGPT 4
	Giuffrè et al. (2024)	Enhances LLM accuracy in digestive diseases via RAG with clinical guidelines	Q&A, Clinical decision-making	European Society of Gastrointestinal Endoscopy (ESGE), European Association for the Study of the Liver (EASL), United European Gastroenterology (UEG) guidelines	Similarity-based retrieval from vector database	None	None	GPT-4 Turbo
Reporting	Bernardi and Cimitile (2024)	Enhanced ALign before fuse model + LLaMA + RAG pipeline	Radiology report generation	MIMIC-CXR	Cosine similarity (top-K reports)	Document Embeddings	None	LLaVa-Med
	Fink et al. (2025)	Enhances GPT-4 with RAG for trauma radiology diagnosis and classification	Diagnostic accuracy	RadioGraphics top ten reading list	Cosine similarity (LlamaIndex)	None	None	GPT-4 Turbo

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Table 4 (*continued*).

Field	Ref.	Main idea	Objective	Dataset	Retrieval technique	Retrieval enhancement		LLM
						Pre-retrieval	Post-retrieval	
Immunology	Chuang et al. (2025)	Predicts IL-6 inducing epitopes using RAG and multi-window CNNs	Epitope prediction	Immune Epitope Database (IEDB)	Vector similarity (Top-5)	None	None	Multi-window CNN
	Sun et al. (2024)	LLM with RAG for detecting severe immune-related adverse events	Diagnostic accuracy	Massachusetts General Hospital EHR data (7555 admissions), Brigham and Women's Hospital EHR data (1270 admissions)	vector database	None	None	Mistral OpenOrca
Nutrition	Zhan et al. (2025)	Combines RAG, multi-task learning, and instruction fine-tuning for DS info extraction	Information extraction (NER, RE, TE, UC)	Clinical Data Repository (CDR)	MedCPT, Contriever, BMRetriever	None	None	Llama2-13B, MedAlpaca-7B
	Soekamto et al. (2025)	Combines VLMs and RAG for personalized recipe recommendations based on ingredient images	Recipe recommendation	Online recipe sources	Search Engine (SE)	None	None	GPT-4o
	Alkhalaf et al. (2024)	Combines zero-shot learning with RAG for EHR summarization and risk factor extraction	Summarization and information extraction	EHR from 40 Australian RACFs	Maximum Marginal Relevance (MMR)	None	None	Llama 2 13B
Pharmacology	X. Zhang et al. (2024)	Pairwise XGBoost + GPT-4 for interpretable bispecific antibody target prediction	Bispecific antibody target pair prediction	Single-cell RNA-seq (NSCLC, Pancreatic, etc.)	Feature database lookup	None	None	GPT-4
	Connor et al. (2025)	Uses RAG framework to enhance drug toxicity predictions by integrating FDA drug-labeling documents.	Drug toxicity assessment	FDA drug-labeling documents	Not specified	None	None	GPT-4
	Wu et al. (2025)	Automates AE extraction from FDA labels using RAG-enhanced LLM	Information retrieval	FDALabel database	Regular database searching	Query Refinement	Advanced Reasoning	Llama 3.1-70B

Table 5
Distribution of RAG architectures by medical field.

Field	Basic	Advanced	Modular
General medicine	4	8	1
Biomedical	0	3	4
Scientific research	1	3	1
Education	1	3	0
Mental health	1	1	2
Gastroenterology	2	3	0
Reporting	1	1	0
Immunology	2	0	0
Nutrition	3	0	0
Pharmacology	2	0	1

Mental health presents a different configuration, with Modular architectures representing 50% of systems, the second highest proportion after biomedicine. In this domain, the explicit separation between entity extraction, dialogue management, and affective ranking enhances conversational stability and contextual alignment, which are central concerns in interactive settings. Gastroenterology, despite 60% Advanced systems, shows no Modular architectures, suggesting that a relatively lightweight Advanced configuration centered on embedding adaptation and document structuring is often sufficient for guideline driven queries. Here, the primary challenge lies in faithfully representing complex documents rather than integrating heterogeneous sources. In contrast, nutrition and immunology remain entirely Basic at 100% Basic systems. This indicates a limited adoption of advanced configurations in fields exposed to implicit or noisy signals, where reliance on similarity alone may not ensure robustness and safety, particularly in the presence of false positives or clinically implicit indicators. Taken together, the cross domain comparison supports a general principle. Modular architectures are most appropriate when the task requires a stable separation between extraction, verification, and generation. Advanced architectures are preferable when the main difficulty concerns input structuring and reasoning control without extensive multiplication of components. Basic architectures remain adequate primarily for bounded queries over highly curated corpora.

Tables 4 and 6 demonstrate that optimal RAG architecture is domain specific and shaped by the dominant clinical objective rather than by a uniform increase in retrieval complexity. In diagnostic and entity relation intensive domains such as biomedicine, the highest performances are achieved when structured knowledge integration is tightly coordinated with reasoning control. SPOKE-based graph integration reaches 95% accuracy on true false biomedical tasks (Soman et al., 2024), and a strategy oriented modular pipeline achieves 93.4% accuracy (Matsumoto et al., 2025), whereas less coordinated graph dependent variants decline to 70.4% (Matsumoto et al., 2024). General medicine shows greater architectural diversity but follows the same pattern. Basic dense similarity improves contextual access, yet substantial gains emerge only when semantic verification, reranking, or structured reasoning are introduced. Advanced configurations reach 70.55 to 82% on benchmark tasks, while prompt reformulation alone remains limited. These findings indicate that for diagnostic workflows requiring explicit validation and mitigation of unsupported conclusions, knowledge grounded Advanced or Modular architectures are more effective than Basic retrieval alone. In contrast, therapeutic domains such as mental health exhibit a fundamentally different optimal configuration. Performance depends less on graph coverage and more on Semantic Tagging, pre-retrieval entity augmentation, and contextual Reranking or Filtering mechanisms that stabilize interaction. Clinical extraction systems achieve 97% F1 with modular entity aware retrieval (Lopez et al., 2025), and sentiment ranked conversational systems reach 82% correction accuracy (Nayinzira and Adda, 2024), whereas similarity based diagnostic style pipelines in psychiatry yield weak case level accuracy of 13 out of 20 correct cases (Gargari et al., 2024). In this context, conversational alignment and affect sensitivity dominate factual graph grounding, and transferring diagnostic style graph enforcement would introduce rigidity without improving safety. A third configuration appears in normative document domains such as gastroenterology and pharmacology, where the dominant failure mode concerns document structure rather than relational ambiguity. Guideline restructuring combined with controlled Prompt Engineering achieves 99% accuracy (Kresevic et al., 2024), while similarity based retrieval with chunk level Filtering achieves only 70% (Fu et al., 2024). Similarly, adverse event extraction from regulatory labels reaches 97.8% F1 through hybrid semantic and exact section level retrieval (Wu et al., 2025), whereas toxicity classification relying mainly on structured prompting remains moderate at 64 to 75% (Connor et al., 2025). In these domains, Data Structuring and hybrid lexical semantic retrieval outperform both graph expansion and isolated reranking because the primary error source lies in section misalignment. Radiology further illustrates that data modality can redefine architectural optimality. Text-grounded Basic retrieval over a curated trauma corpus achieves 100% diagnostic accuracy within a constrained classification benchmark (Fink et al., 2025). However, multimodal report generation addresses the more complex task of translating visual findings into structured clinical narratives. Image–text alignment mechanisms achieve 49% RadGraph F1 (Bernardi and Cimitile, 2024), reflecting moderate structural fidelity. While Basic retrieval suffices for narrow diagnostic classification, multimodal alignment remains a necessary and promising direction for realistic radiology reporting. Finally, nutrition and immunology remain dominated by Basic pipelines and demonstrate strong metrics under explicit conditions but fragility under ambiguity. Nutrition reaches 93.74% F1 in supplement relation extraction and 99.25% accuracy for explicit record summarization (Zhan et al., 2025; Alkhalaf et al., 2024), yet performance declines when implicit cues are involved. Immunology achieves 98.1% sensitivity in immune event detection and 90.74 AUC in epitope prediction (Sun et al., 2024; Chuang et al., 2025), but reduced positive predictive value reveals insufficient false positive control. Across domains, the comparative evidence supports a clear operational conclusion. Knowledge graph enhanced Advanced or Modular architectures are optimal for structured diagnostic reasoning. Semantic Tagging combined with contextual Reranking is optimal for therapeutic interaction. Data Structuring with hybrid section level retrieval is optimal for guideline and regulatory corpora. Multimodal pre-retrieval analysis is essential when visual fidelity governs correctness. Basic similarity retrieval is sufficient only for bounded queries over highly curated corpora and is inadequate for relational, ambiguous, or safety critical tasks. Architectural refinement must therefore align with the dominant clinical error mode rather than follow a uniform increase in retrieval sophistication. Despite this, advanced refinements and complex task settings remain insufficiently explored in several domains, revealing a gap between demonstrated potential and current practice.

As summarized in Table 6, RAG technology enhances LLM capabilities across various medical specialties or fields by grounding responses in authoritative knowledge. In mental health practice, RAG-enhanced tools provide clinicians with instant access to diagnostic criteria and therapeutic frameworks, significantly improving the quality of patient assessments. Surgical teams benefit from real-time retrieval of procedural guidelines and case studies, enhancing both decision-making and patient safety. For complex diagnosis of rare diseases, RAG systems synthesize information from diverse medical literature and databases, providing clinicians with comprehensive diagnostic support. Medical education has been particularly transformed, with RAG-powered tutoring systems delivering curriculum-perfect explanations and case examples. In pharmaceutical applications, these systems enable the rapid identification of drug interactions and adverse effects by processing a large amount of regulatory documentation. Immunology researchers leverage RAG to navigate complex biomarker relationships and clinical trial data.

4.11.2. Clinical validity and ethical implications

In healthcare, clinical validity is inseparable from ethical responsibility. A RAG system that achieves strong quantitative performance yet produces clinically misleading, incomplete, or unsafe outputs poses di-

rect patient risk. As highlighted in prior healthcare reviews (Amugongo et al., 2025; Neha et al., 2025), correctness, safety, bias mitigation, and privacy protection are not secondary considerations but foundational requirements for responsible deployment.

Across the reviewed domains, limitations in evaluation frameworks translate into ethical vulnerabilities. Diagnostic systems reporting high accuracy may not account for error severity or real-world prevalence shifts. Immune event detection with high sensitivity but low positive predictive value (Sun et al., 2024) illustrates how false-positive burden can strain clinical workflow and undermine safety despite strong aggregate metrics. Radiology report generation demonstrates that structural alignment metrics are more clinically meaningful than lexical similarity (Bernardi and Cimitile, 2024), yet even these do not measure downstream diagnostic consequences. In mental health applications, fluency and semantic similarity fail to capture therapeutic appropriateness or psychological safety, while GPT-based judging may overestimate helpfulness without calibrated harm assessment (Sung et al., 2024).

Privacy and governance risks are similarly underreported. Many reviewed systems operate over electronic health records or institution-specific corpora (Alkhalaf et al., 2024; Sun et al., 2024; Myers et al., 2025), yet few explicitly document de-identification protocols, access control mechanisms, or retrieval-layer safeguards. As noted in healthcare-focused analyses (Neha et al., 2025), dense retrievers without strict governance may expose protected health information through indexing artifacts or metadata leakage. Moreover, major regulatory frameworks, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States or the General Data Protection Regulation (GDPR) in Europe, are named in only a handful of the reviewed studies, and fewer still discuss concrete alignment or compliance measures, despite their direct relevance to the handling of clinical data in RAG systems. The introduction of retrieval components expands the attack surface, requiring secure indexing, controlled retrieval, context filtering, and auditability.

Bias and equity considerations remain largely unexplored in the reviewed studies. Although grounding responses through retrieval reduces hallucination, it does not eliminate biases embedded in language models or training corpora (Amugongo et al., 2025). Few systems systematically evaluate demographic fairness, cross-institution generalization, or language disparities. This omission is ethically significant, particularly in diagnostic or therapeutic contexts where unequal performance may disproportionately affect vulnerable populations. An examination of the datasets summarized in Table 4 shows that only 6 of the 57 studies draw on explicitly non-Western or non-English-language sources, including Chinese, Japanese, and Taiwanese institutions, while the remainder rely predominantly on United States, European, or international English-language databases and guidelines. Similarly, only 4 studies explicitly validate across more than one institution or site, with the rest relying on a single dataset or source.

Safety evaluation is often limited to accuracy reporting rather than structured harm analysis. Only a minority of studies incorporate adversarial testing, failure-case categorization, or explicit safety flags. Yet harm modes differ substantially across specialties: diagnostic misclassification in radiology, guideline misinterpretation in gastroenterology, adverse event mislabeling in pharmacology, and inappropriate affective responses in mental health. Ethical robustness therefore requires domain-sensitive risk modeling rather than uniform metric reporting.

Finally, accountability and regulatory alignment remain emerging challenges. As RAG systems increasingly influence clinical workflows, questions of liability, human oversight, and explainability become central. Transparent source traceability, clear documentation of retrieval corpora, and explicit delineation of human-in-the-loop control are necessary to align medical RAG systems with evolving governance frameworks (Abo El-Enen et al., 2025).

In summary, clinical validity cannot be reduced to automated metric optimization. Ethical deployment demands calibrated evaluation practices that incorporate safety analysis, bias assessment, privacy safeguards, and domain-specific risk modeling. The absence of structured, field-level validation standards constitutes a significant gap in the current literature. Addressing this gap through harmonized benchmarking, transparent reporting, and specialty-aware safety evaluation represents a necessary step toward trustworthy integration of Retrieval-Augmented Generation in healthcare.

In this section, we have described the applications and benefits of RAGs in several medical fields and their limits, with a discussion about their clinical validity, safety and ethical implications. However, there are still some potential improvements that could be explored in future work, which will be discussed in the next section.

5. Concluding remarks and future research directions

This comprehensive survey provides a systematic examination of RAG systems across various healthcare domains, revealing critical insights into their current implementations and effectiveness. Through rigorous analysis of high-quality studies, we demonstrate that clinical utility is fundamentally determined by how well RAG architectures align with specialty-specific requirements and workflows. Our analysis suggests that deterministic retrieval over curated knowledge bases is particularly effective in highly structured diagnostic settings such as radiology and guideline-driven surgical decision support. In contrast, conversational applications such as mental health require more flexible frameworks capable of handling contextual and emotional nuances. However, retrieval enhancement techniques and modular architectures are underexplored and less leveraged for specific domains while many studies employ a basic RAG.

The study identifies several persistent challenges that currently limit the broader clinical translation of RAG systems. These include maintaining knowledge freshness in rapidly evolving medical fields, addressing corpus incompleteness in niche specialties, and resolving inconsistencies in evaluation frameworks that hinder cross-domain comparisons. We observe significant variations in system designs, evaluation metrics, and performance outcomes across different medical fields, highlighting the need for more standardized but flexible approaches. Moreover, the most commonly used metrics lack the precision needed to accurately assess a model's performance for specific medical tasks. Thus, current implementations struggle to balance technical performance with practical workflow integration, often prioritizing algorithmic metrics over clinically meaningful outcomes. These findings collectively underscore the complex interplay between technological capabilities and healthcare requirements that characterize medical RAG systems.

This review has several limitations. First, substantial heterogeneity in evaluation metrics, datasets, and reporting standards across studies limits direct cross specialty comparison and prevents normalized performance assessment. Second, although underexplored medical domains are identified, the review does not develop detailed technical implementation pathways for deploying RAG architectures in those specialties. Third, while architectural patterns are analyzed in depth, data centric dimensions such as corpus construction practices, knowledge base completeness, and dataset governance considerations are not systematically examined as an independent analytical layer. Assessing demographic fairness and dataset governance across the corpus remains an important direction for future work. Similarly, computational cost, hardware requirements, and real-world implementation remain largely unreported across the reviewed literature. Together with the broader deployment and human-centered evaluation gaps in RAG healthcare research, these would be better addressed by a focused review of its own.

For future improvement of RAG implementations in healthcare, three key research directions emerge as particularly critical. First, there is an urgent need to develop standardized evaluation protocols that meaningfully capture both technical and clinical performance metrics. Generic metrics like accuracy, BLEU, and ROUGE are inadequate for

Table 6
RAG contributions to medical fields.

Field	Ref.	LLM	Evaluation	Benefits
General medicine	Ong et al. (2024)	Llama 2/3	Not specified	Reduces surgical errors through guideline retrieval
	Fuellen et al. (2025)	GPT-4, Claude Opus	Assessment by another model: - Good improvements of GPT4 in: Correct, Holistic, Longitudinal, Aging-related - Moderate improvement of GPT4 in: Longitudinal - Good improvement of Claude Opus in: Holistic	Enhances geroprotective intervention analysis with knowledge graph verification
	Gilbert et al. (2024)	Custom LLM framework	Not specified	Prevents medical misinformation via SNOMED-CT grounded outputs
	Su et al. (2025)	LLaVA-Med	Results: - Zero-shot accuracy: 49.4 - RAI: 0.98 - SS: 0.93 - Relative LLM scores: 0.96	Secures Internet of Medical Things (IoMT) data management through multimodal retrieval
	Zhang et al. (2023)	Flan-T5	Accuracy: - MedQA-USMLE: 38.15%–48.94% - StrategyQA: 56.57%–70.55%	Empowers small LMs for clinical QA with external knowledge
	Yang et al. (2024)	GPT-3.5-turbo	Results: - Context precision: 0.87 - Context recall: 0.89 - Faithfulness: 0.65 - Answer relevancy: 0.73 - Answer correctness: 0.75	Improves rare disease diagnosis via multi-source fusion
	Zheng et al. (2025)	GPT-4o	Results: - Precision: 94% - Recall: 97% - F1-score: 95%	Optimizes physician–patient matching through consultation record analysis
	Hu et al. (2025)	Series of GPT, Claude, LLaMA, and Mistral.	Proprietary models: - Answer correctness (gpt-3.5-turbo): 81% - Answer relevancy (gpt-4o): 99% - Answer similarity (gpt-3.5-turbo): 99% - Context precision (gpt-4-turbo): 90% - Context recall (claude-3-opus): 98% - Faithfulness (claude-3-haiku): 98% Open-source models: - Answer correctness (mistral-8 × 22b): 90.3% - Answer relevancy (mistral-7b): 96% - Answer similarity (mistral-8 × 22b): 92.8% - Context precision (mistral-nemo): 88.9% - Context recall (llama2-7b): 93.3% - Faithfulness (Mistral-large-2): 97.7%	Improves customs clearance through multimodal document parsing
	Myers et al. (2025)	Not specified	Mean Average Precision (MAP): - UW dataset: 40.3% - MIMIC dataset: 47.5%	Improves EHR retrieval through optimal embedding strategies
	Glicksberg et al. (2024)	GPT-4	Results: - AUC: 87.4% - Accuracy: 83.1% - AUPRC: 70.9% - F1-score: 54.6%	Enhances emergency department admission prediction with clinical note analysis
	Zhang et al. (2025)	GPT-4	Results: - Accuracy: 82% - Precision: 62% - Recall: 65% - F1-score: 61%	Standardizes forensic injury appraisal with judicial knowledge retrieval
	Nahass et al. (2024)	GPT-3.5-Turbo	Correct response rate: 54.3% Correct reasoning: 58.07%	Enhances plastic surgery exam preparation through curated content retrieval
	Zelin et al. (2024)	GPT 3.5	Accuracy: 43%	Improves rare disease diagnosis accuracy by retrieving phenotypic information
	Yazaki et al. (2024)	GPT 3.5	- Correct triage rate: 70% - Under-triage rate: 8%	Improves emergency patient triage decision support through guideline-grounded retrieval
Biomedicine	Matsumoto et al. (2024)	BioGPT	Accuracy: - multiple choice questions: 70.4% (1-hop reasoning) and 71.8% (2-hop) - true or false questions: 80.3% (1-hop) and 62.9% (2-hop).	Advances biomedical research via knowledge graph traversal

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Table 6 (*continued*).

Field	Ref.	LLM	Evaluation	Benefits
	Matsumoto et al. (2025)	GPT-4o Mini	Exact match (1-hop): - True/false: 90.9% - Multiple choice: 93.4% Overlap (1-hop): - Open-ended: 88.4%	Enhances clinical reasoning with dynamic knowledge integration
	Cinquin (2024)	GPT-4o	Improve factual accuracy and instruction following	Prevents bioinformatics errors through API-corrected queries
	Jeong et al. (2024)	LLaMA2 (fine-tuned)	Accuracy: 46.5% LiveQA: - Rouge-1: 19.7 - Rouge-2: 3.1 - Rouge-L: 13.4 - BERTScore: 77.2 MedicationQA: - Rouge-1: 17.6 - Rouge-2: 3.3 - Rouge-L: 13.5 - BERTScore: 80.2	Enables self-reflective medical QA with retrieval-augmented validation
	Kim and Yoon (2024)	GPT-3.5	BLEU: 79.15% ROUGE-1: - Recall: 89.66% - Precision: 92.86% - F1-score: 91.23% ROUGE-2: - Recall: 86.44% - Precision: 82.26% - F1-score: 84.30% ROUGE-L: - Recall: 86.21% - Precision: 89.29% - F1-score: 87.72%	Accelerates drug discovery through biomolecular interaction extraction
	Perkins et al. (2025)	GPT-4, Claude3	Accuracy: - GPT-4: 90% (63/70) - Claude3: 93% (65/70)	Improves microbiology QA through clinical manual retrieval
	Di Maria et al. (2024)	gpt- 3.5-turbo-instruct	Accuracy: 87% (87/100) BC2GM dataset: - Precision: 57% - Recall: 43% - F1-score: 49% NCBI disease dataset: - Precision: 87% - Recall: 60% - F1-score: 71%	Visualizes biomedical knowledge through on-demand graph construction
	Soman et al. (2024)	GPT-3.5-Turbo	Accuracy: - True/False dataset: 95% - MCQ dataset: 79%	Optimizes disease-centric QA via context pruning
Scientific research	Wysocka et al. (2024)	GPT-4	Factual answers: - 70% for chemical entity recognition - 43.6% for chemical-fungi relations Hallucination: 54.4%	Reduces expert evaluation workload in antibiotic research
	Luu and Buehler (2023)	Orca-2-13b	Score: 90%	Inspires biomimetic materials design through biological pattern retrieval
	Kurland et al. (2025)	GPT-4	Manual review of factual and citation accuracy: 97.5%	Automates literature synthesis with citation-accurate summarization
	Li et al. (2024)	GPT-4 Turbo	Evaluation by experts with scoring scale of 5 for each dimension: Literature recommendation task: - Relevance: 4.40 - Quality: 4.01 Summarization task: - Accuracy: 4.40 - Comprehensiveness: 3.40 - Reference integration: 4.60	Combats research misinformation through peer-reviewed source retrieval
	Craig and Drăghici (2024)	GPT-4o	Not specified	Bridges experimental data with relevant literature
Education	Fang et al. (2025)	InternLM2-Chat-7B, Baichuan2-7B-Chat, Qwen-7b-Chat	Mean scores: Content accuracy (4.13), Depth and detail (3.70), Practicality and innovation (3.30), Logic and organization (4.50)	Personalizes teacher training through scenario retrieval

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Table 6 (*continued*).

Field	Ref.	LLM	Evaluation	Benefits
	Arun et al. (2024)	ChatGPT 3.5	Accuracy: 77.8% (14/18) Likert scale mean: - Factual accuracy: 4.78 - Relevance: 4.39 - Completeness: 4.33 - Coherence: 4.78 - Fluency: 5.00	Preserves anatomy education integrity with textbook-grounded answers
	Chang and Lin (2024)	GPT-4.0	Accuracy: 69.81%	Democratizes medical education via bilingual exam preparation
	Zhao et al. (2025)	Custom GPT (based on ChatGPT-4)	Accuracy: 90%	Enhances nursing exam preparation through curriculum-aligned retrieval
Mental health	Lopez et al. (2025)	GPT-4, Flan-T5, Med42, Llama-3, Mistral	Results: - Information extraction F1-score (GPT-4): 97% - Inference time: 4.95 s - Reduction in token usage: >70%	Enables precise clinical variable extraction from EHRs
	Sung et al. (2024)	Mistral-7B	Score of GPT-4 (with a scale of 1 to 10 for each question): 726/800	Solves psychiatry data scarcity through dynamic dataset generation
	Gargari et al. (2024)	GPT-3.5 Turbo, Nemotron-3-8B	GPT-3.5: 13/20 correct Nemotron: 7/20 correct	Improves psychiatric diagnosis with DSM-5 case retrieval
	Nayinzira and Adda (2024)	mistral-large-latest	Results: - Answer correctness: 0.82 - Answer relevancy: 0.80 - Token usage: 3701 - Latency: 43.22	Enhances mental health support via sentiment-aware counseling retrieval
Gastroenterology	Fu et al. (2024)	gpt-3.5-turbo-16k	Accuracy: 70% Level of detail: 94.5%	Reduces nursing workload with instant endoscopic care answers
	Zhou et al. (2024)	gpt-3.5-turbo	Context recall rate: 95%. faithfulness to the source: 93.73%. Relevance of answers: 92.28%	Localizes gastrointestinal knowledge for Chinese clinical practice
	Kresevic et al. (2024)	GPT-4 Turbo	Accuracy: 99%	Prevents HCV guideline misinterpretations
	Malik et al. (2024)	ChatGPT 4	Scores by investigators (Likert scale 1..5): - Relevance: 4.01 - Accuracy: 3.67 - Clarty: 4.04	Improves acute liver failure management with American College of Gastroenterology-grounded advice
	Giuffrè et al. (2024)	GPT-4 Turbo	Accuracy: 90%	Maintains digestive disease guideline compliance
Reporting	Bernardi and Cimitile (2024)	LLaVa-Med	Lexical/Semantic metrics: - BLEU: 0.43 - GLEU: 0.44 - METEOR: 0.38 - BERTscore (macro F1): 0.66 Domain-specific medical imaging metrics: - RadCliQ F1 (lower is better): 0.17 - CheXbert F1: 0.45 - RadGraph F1: 0.49	Reduces radiologist workload with auto-generated reports
	Fink et al. (2025)	GPT-4 Turbo	Subjective rating by expert (for trustworthiness and quality of explanation of responses). Diagnosis accuracy: - Correct (score 5): 100% Classification: - Correct (score 5): 96% - Close (score 4): 2% - Wrong (score 1–3): 2%	Prevents trauma misclassification with reference retrieval
Immunology	Chuang et al. (2025)	Multi-window CNN	BACC: 0.8587 AUC: 0.9074 MCC: 0.6407 Sensitivity: 0.7808 Specificity: 0.9365 Accuracy: 0.9195	Predicts cytokine storm risks through epitope retrieval
	Sun et al. (2024)	Mistral OpenOrca	Results: - Sensitivity: 98.1% - Specificity: 95.7% - Positive predictive value: 17.1% - Negative predictive value: 100%"	Accelerates immunotoxicity detection in EHRs with high sensitivity

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Table 6 (*continued*).

Field	Ref.	LLM	Evaluation	Benefits
Nutrition	Zhan et al. (2025)	Llama2-13B, MedAlpaca-7B	F1-score: - Named entity recognition (Llama2-13B): 87.39% - Relation extraction (Llama2-13B): 93.74% - Triple extraction (Llama2-7B): 79.45% - Usage classification (MedAlpaca-7B): 93.45%	Standardizes dietary supplement documentation from clinical records
	Soekamto et al. (2025)	GPT-4o	Precision: 83% Recall: 91% Accuracy: 77% F1-score: 86%	Personalizes nutrition through image-based recipe retrieval
	Alkhalaf et al. (2024)	Llama 2 13B	Accuracy: - Summarization: 99.25% - Risk factors extraction: 90%	Automates malnutrition risk detection in aged care records
Pharmacology	X. Zhang et al. (2024)	GPT-4	AUC: 89.29%	Accelerates bispecific antibody development
	Connor et al. (2025)	GPT-4	Per-organ accuracy: - Kidney: 72% - Heart: 64% - Liver: 75%	Enhances drug toxicity assessment with FDA label retrieval
	Wu et al. (2025)	Llama 3.1-70B	Drug-induced liver injury classification: - Accuracy: 96.7% - Recall: 99.4% - Precision: 96.2% - Specificity: 89.5% - F1-score: 97.8% Drug-induced cardiotoxicity classification: 93.1% - Accuracy: 90.1% - Recall: 94.1% - Precision: 92.1% - Specificity: 80.2% - F1-score: 93.1% Drug adverse event term recognition: 91.1% - Precision: 90.6% - Recall: 93.9% - F1-score: 91.1%	Automates adverse event monitoring from drug labels

clinical deployment. To ensure safety and efficacy, future work must adopt domain-specific evaluation frameworks that directly measure clinical utility, including therapeutic alliance in mental health, diagnostic concordance in radiology, guideline adherence in decision support, learning outcomes in education, and adverse event validation in pharmacology. Concretely, this could take the form of standardized reporting templates specifying minimum required metrics for each clinical task type, together with shared benchmark datasets curated with clinician input for every medical field. This shift toward clinically grounded evaluation is essential for developing trustworthy RAG systems in healthcare. Second, innovative approaches are needed to develop adaptive retrieval mechanisms capable of integrating continuously evolving medical knowledge while maintaining accuracy, consistency, and traceability. Modular and adaptive architectures are promising in this context because they allow flexible updating of knowledge components without redesigning the entire system. This could be realized, for example, through versioned or incrementally updatable knowledge bases and retriever components that can be refreshed independently of the generation module. Third, the field must focus on designing lightweight, optimized architectures that can meet the specific needs of different clinical workflows without excessive computational overhead. A well-designed modular and adaptive architecture can help mitigate computational overhead and latency that hinder real-time clinical deployment. By dynamically adapting to task complexity and selectively activating components, modular systems can better balance reasoning depth with response time, ensuring performance remains compatible with time-sensitive medical environments. Progress in these areas will require deeper collaboration between clinicians, computer scientists, and domain experts to ensure solutions address real clinical needs while maintaining rigorous standards of safety, efficacy, and ethical responsibility. By addressing these challenges, the next generation of medical RAG systems can transition from experimental tools to reliable components of healthcare infrastructure that genuinely enhance clinical decision-making and patient care.

Finally, we stress the fact that several important medical fields were not covered by our survey. This simply means that the use of RAGs in these domains is not explored yet despite their important. Among the top 15 medical fields based on research contributions, immunology is ranked first, followed by neurology, ophthalmology, musculoskeletal, and dermatology (Mikulic, 2025). Our paper covered immunology as a medical field where RAGs were used, but not the rest of the previously mentioned fields. This reflects insufficient eligible studies at the time of our search. We recommend future reviews revisit these fields as RAG research expands into them. These findings mean that there is still considerable work ahead in the use of RAGs in healthcare, and this review aims to support that progress by giving research scientists and healthcare practitioners a clearer picture of the current architectural landscape and its fit to specific clinical needs.

CRediT authorship contribution statement

Mohamed Yassine Landolsi: Writing – review & editing, Writing – original draft, Visualization, Methodology. **Imen Fadhli:** Writing – review & editing, Writing – original draft, Methodology. **Amine Boufaied:** Validation, Supervision, Methodology. **Lotfi Ben Romdhane:** Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

See Tables 7 and 8.

Table 7
Methodological quality assessment of the included empirical studies using the adapted MMAT framework.

Field	Ref.	Q1	Q2	Q3	Q4	Q5	Total	Quality
Biomedicine	Soman et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Matsumoto et al. (2025)	1	0.5	0	0.5	0	2	LQ
	Jeong et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Matsumoto et al. (2024)	1	1	0.5	0.5	0	3	MQ
	Di Maria et al. (2024)	1	1	0.5	0.5	0	3	MQ
	Perkins et al. (2025)	1	1	0.5	0.5	0.5	3.5	MQ
	Cinquin (2024)	1	1	0.5	0	0.5	3	MQ
	Kim and Yoon (2024)	1	1	0.5	0.5	0	3	MQ
Education	Chang and Lin (2024)	1	1	0.5	1	0.5	4	HQ
	Arun et al. (2024)	1	0.5	0.5	1	0	3	MQ
	Zhao et al. (2025)	1	0.5	1	1	0	3.5	MQ
	Fang et al. (2025)	1	1	0.5	1	1	4.5	HQ
Gastroenterology	Fu et al. (2024)	1	0.5	1	1	0	3.5	MQ
	Malik et al. (2024)	1	0.5	0.5	1	0	3	MQ
	Zhou et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Kresevic et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Giuffrè et al. (2024)	1	0.5	0	0.5	0.5	2.5	LQ
General medicine	Gilbert et al. (2024)	1	0.5	0	0.5	1	3	MQ
	Yazaki et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Glicksberg et al. (2024)	1	1	1	1	0	4	HQ
	Su et al. (2025)	1	1	0	0.5	0.5	3	MQ
	Hu et al. (2025)	1	1	0	1	0.5	3.5	MQ
	Zhang et al. (2025)	1	1	1	1	0	4	HQ
	Zheng et al. (2025)	1	1	1	1	1	5	HQ
	Zhang et al. (2023)	1	1	1	1	0	4	HQ
	Myers et al. (2025)	1	1	0.5	1	0	3.5	MQ
	Nahass et al. (2024)	1	1	0.5	1	0.5	4	HQ
	Zelin et al. (2024)	1	1	0.5	1	0	3.5	MQ
	Yang et al. (2024)	1	1	0	0.5	0.5	3	MQ
	Ong et al. (2024)	1	1	0	0.5	0.5	3	MQ
	Fuellen et al. (2025)	0.5	0.5	0.5	0.5	0.5	2.5	LQ
Immunology	Sun et al. (2024)	1	1	1	1	0	4	HQ
	Chuang et al. (2025)	1	1	0.5	0.5	0	3	MQ
Mental health	Sung et al. (2024)	1	1	1	1	0.5	4.5	HQ
	Lopez et al. (2025)	1	1	1	1	0	4	HQ
	Gargari et al. (2024)	1	0.5	0.5	0.5	0	2.5	LQ
	Nayinzira and Adda (2024)	1	1	0.5	0.5	0	3	MQ
Nutrition	Alkhalaf et al. (2024)	1	1	1	1	0.5	4.5	HQ
	Soekamto et al. (2025)	1	1	0.5	1	0	3.5	MQ
	Zhan et al. (2025)	1	1	0.5	1	0	3.5	MQ
Pharmacology	X. Zhang et al. (2024)	1	1	1	1	0.5	4.5	HQ
	Connor et al. (2025)	1	0.5	0.5	1	0.5	3.5	MQ
	Wu et al. (2025)	1	1	0	1	0.5	3.5	MQ
Reporting	Bernardi and Cimitile (2024)	1	1	0.5	1	0.5	4	HQ
Scientific research	Kurland et al. (2025)	1	1	0	1	0.5	3.5	MQ
	Luu and Buehler (2023)	1	1	0	1	0.5	3.5	MQ
	Wysocka et al. (2024)	1	1	0	1	0.5	3.5	MQ
	Craig and Drăghici (2024)	1	1	0.5	0	0.5	3	MQ
	Li et al. (2024)	1	1	0	1	0.5	3.5	MQ

Table 8
Studies included in the survey.

Title	Year	Venue	DOI	Reason for inclusion
Steering veridical large language model analyses by correcting and enriching generated database queries: first steps toward ChatGPT bioinformatics	2025	Briefings in Bioinformatics	10.1093/bib/bbaf045	Eligible field (≥3 studies): biomedical + Q1 journal
Improving large language model applications in biomedicine with retrieval-augmented generation: a systematic review, meta-analysis, and clinical development guidelines	2025	Journal of the American Medical Informatics Association	10.1093/jamia/ocaf008	Eligible field (≥3 studies): biomedical + Q1 journal
ESCARGOT: an AI agent leveraging large language models, dynamic graph of thoughts, and biomedical KGs for enhanced reasoning	2025	Bioinformatics	10.1093/bioinformatics/btaf031	Eligible field (≥3 studies): biomedical + Q1 journal
KRAGEN: a KG-enhanced RAG framework for biomedical problem solving using large language models	2024	Bioinformatics	10.1093/bioinformatics/btae353	Eligible field (≥3 studies): biomedical + Q1 journal
Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models	2024	Bioinformatics	10.1093/bioinformatics/btae238	Eligible field (≥3 studies): biomedical + Q1 journal
Biomedical KG-optimized prompt generation for large language models	2024	Bioinformatics	10.1093/bioinformatics/btae560	Eligible field (≥3 studies): biomedical + Q1 journal
NetMe 2.0: a web-based platform for extracting and modeling knowledge from biomedical literature as a labeled graph	2024	Bioinformatics	10.1093/bioinformatics/btae194	Eligible field (≥3 studies): biomedical + Q1 journal
Retrieval-augmented generation salvages poor performance from large language models in answering microbiology-specific multiple-choice questions	2025	Journal of Clinical Microbiology	10.1128/jcm.01624-24	Eligible field (≥3 studies): biomedical + Q1 journal
The Sociodemographic Biases in Machine Learning Algorithms: A Biomedical Informatics Perspective	2024	Life	10.3390/life14060652	Eligible field (≥3 studies): biomedical + Q1 journal
VAIV bio-discovery service using transformer model and retrieval augmented generation	2024	BMC bioinformatics	10.1186/s12859-024-05903-6	Eligible field (≥3 studies): biomedical + Q1 journal
ChatGPT versus a customized AI chatbot (Anatbuddy) for anatomy education: A comparative pilot study	2024	Anatomical Sciences Education	10.1002/ase.2502	Eligible field (≥3 studies): education + Q1 journal
Evaluating simulated teaching audio for teacher trainees using RAG and local LLMs	2025	Scientific Reports	10.1038/s41598-025-87898-5	Eligible field (≥3 studies): education + Q1 journal
Deriving insights from enhanced accuracy: Leveraging prompt engineering in custom GPT for assessing Chinese Nursing Licensing Exam	2025	Nurse Education in Practice	10.1016/j.nepr.2025.104284	Eligible field (≥3 studies): education + Q1 journal
203 Enhancing Artificial Intelligence Performance in Bilingual Emergency Medicine Board Exam: The Impact of Retrieval-Augmented Generation	2024	Annals of Emergency Medicine	10.1016/j.annemergmed.2024.08.208	Eligible field (≥3 studies): education + CORE B conference
Optimization of hepatological clinical guidelines interpretation by large language models: a retrieval augmented generation-based framework	2024	npj Digital Medicine	10.1038/s41746-024-01091-y	Eligible field (≥3 studies): gastroenterology + Q1 journal
Optimizing large language models in digestive disease: strategies and challenges to improve clinical outcomes	2024	Liver International	10.1111/liv.16039	Eligible field (≥3 studies): gastroenterology + Q1 journal
Evaluating Artificial Intelligence-Driven Responses to Acute Liver Failure Queries: A Comparative Analysis Across Accuracy, Clarity, and Relevance	2024	American Journal of Gastroenterology	10.14309/ajg.0000000000003255	Eligible field (≥3 studies): gastroenterology + Q1 journal
GastroBot: a Chinese gastrointestinal disease chatbot based on the retrieval-augmented generation	2024	Frontiers in Medicine	10.3389/fmed.2024.1392555	Eligible field (≥3 studies): gastroenterology + Q1 journal
Application of large language model combined with retrieval enhanced generation technology in digestive endoscopic nursing	2024	Frontiers in Medicine	10.3389/fmed.2024.1500258	Eligible field (≥3 studies): gastroenterology + Q1 journal
Validation requirements for AI-based intervention-evaluation in aging and longevity research and practice	2024	Ageing Research Reviews	10.1016/j.arr.2024.102617	Eligible field (≥3 studies): general medicine + Q1 journal
Augmented non-hallucinating large language models as medical information curators	2024	npj Digital Medicine	10.1038/s41746-024-01081-0	Eligible field (≥3 studies): general medicine + Q1 journal
SurgeryLLM: a retrieval-augmented generation large language model framework for surgical decision support and workflow enhancement	2024	npj Digital Medicine	10.1038/s41746-024-01391-3	Eligible field (≥3 studies): general medicine + Q1 journal
Hybrid RAG-Empowered Multi-Modal LLM for Secure Data Management in Internet of Medical Things: A Diffusion-Based Contract Approach	2025	IEEE Internet of Things Journal	10.1109/JIOT.2024.3521425	Eligible field (≥3 studies): general medicine + Q1 journal

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Table 8 (*continued*).

Title	Year	Venue	DOI	Reason for inclusion
RDguru: A Conversational Intelligent Agent for Rare Diseases	2024	IEEE Journal of Biomedical and Health Informatics	10.1109/JBHI.2024.3464555	Eligible field (≥3 studies): general medicine + Q1 journal
Evaluating the accuracy of a state-of-the-art large language model for prediction of admissions from the emergency room	2024	Journal of the American Medical Informatics Association	10.1093/jamia/ocae103	Eligible field (≥3 studies): general medicine + Q1 journal
Lessons learned on information retrieval in electronic health records: a comparison of embedding models and pooling strategies	2025	Journal of the American Medical Informatics Association	10.1093/jamia/ocae308	Eligible field (≥3 studies): general medicine + Q1 journal
Rare disease diagnosis using knowledge guided retrieval augmentation for ChatGPT	2024	Journal of Biomedical Informatics	10.1016/j.jbi.2024.104702	Eligible field (≥3 studies): general medicine + Q1 journal
ICCA-RAG: Intelligent Customs Clearance Assistant Using Retrieval-Augmented Generation (RAG)	2025	IEEE Access	10.1109/ACCESS.2025.3544408	Eligible field (≥3 studies): general medicine + Q1 journal
Integrating retrieval-augmented generation for enhanced personalized physician recommendations in web-based medical services: model development study	2025	Frontiers in Public Health	10.3389/fpubh.2025.1501408	Eligible field (≥3 studies): general medicine + Q1 journal
Large Language Models Applied to Health Care Tasks May Improve Clinical Efficiency, Value of Care Rendered, Research, and Medical Education	2025	Arthroscopy	10.1016/j.arthro.2024.12.010	Eligible field (≥3 studies): general medicine + Q1 journal
Emergency Patient Triage Improvement through a Retrieval-Augmented Generation Enhanced Large-Scale Language Model	2024	Prehospital Emergency Care	10.1080/10903127.2024.2374400	Eligible field (≥3 studies): general medicine + Q1 journal
Injury degree appraisal of large language model based on retrieval-augmented generation and deep learning	2025	International Journal of Law and Psychiatry	10.1016/j.ijlp.2025.102070	Eligible field (≥3 studies): general medicine + Q1 journal
Prompt engineering to increase GPT3.5's performance on the Plastic Surgery In-Service Exams	2024	Journal of Plastic, Reconstructive & Aesthetic Surgery	10.1016/j.bjps.2024.09.001	Eligible field (≥3 studies): general medicine + Q1 journal
Knowledge-augmented reasoning distillation for small language models in knowledge-intensive tasks	2023	Advances in Neural Information Processing Systems	10.5555/3666122.3668231	Eligible field (≥3 studies): general medicine + CORE A* conference
Enhancing Precision in Detecting Severe Immune-Related Adverse Events: Comparative Analysis of Large Language Models and International Classification of Disease Codes in Patient Records	2024	Journal of Clinical Oncology	10.1200/JCO.24.00326	Eligible field (≥3 studies): immunology + Q1 journal
RAG_MCNNIL6: A Retrieval-Augmented Multi-Window Convolutional Network for Accurate Prediction of IL-6 Inducing Epitopes	2025	Journal of Chemical Information and Modeling	10.1021/acs.jcim.4c02144	Eligible field (≥3 studies): immunology + Q1 journal
Evaluating the role of large language models in inflammatory bowel disease patient information	2024	World Journal of Gastroenterology	10.3748/wjg.v30.i29.3538	Eligible field (≥3 studies): immunology + Q1 journal
Diagnostic accuracy of large language models in psychiatry	2024	Asian Journal of Psychiatry	10.1016/j.ajp.2024.104168	Eligible field (≥3 studies): mental health + Q1 journal
The revolution of generative artificial intelligence in psychology: The interweaving of behavior, consciousness, and ethics	2024	Acta Psychologica	10.1016/j.actpsy.2024.104593	Eligible field (≥3 studies): mental health + Q1 journal
Clinical entity augmented retrieval for clinical information extraction	2025	npj Digital Medicine	10.1038/s41746-024-01377-1	Eligible field (≥3 studies): mental health + CORE A* conference
A New Pipeline for Generating Instruction Dataset via RAG and Self Fine-Tuning	2024	2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)	10.1109/compsac61105.2024.00371	Eligible field (≥3 studies): mental health + CORE B conference
SentimentCareBot: Retrieval-Augmented Generation Chatbot for Mental Health Support with Sentiment Analysis	2024	Procedia Computer Science	10.1016/j.procs.2024.11.118	Eligible field (≥3 studies): mental health + CORE B conference
RAMIE: retrieval-augmented multi-task information extraction with large language models on dietary supplements	2025	Journal of the American Medical Informatics Association	10.1093/jamia/ocaf002	Eligible field (≥3 studies): nutrition + Q1 journal
Applying generative AI with retrieval augmented generation to summarize and extract key clinical information from electronic health records	2024	Journal of Biomedical Informatics	10.1016/j.jbi.2024.104662	Eligible field (≥3 studies): nutrition + Q1 journal
Pic2Plate: A Vision-Language and Retrieval-Augmented Framework for Personalized Recipe Recommendations	2025	Sensors	10.3390/s25020449	Eligible field (≥3 studies): nutrition + Q1 journal
Is ChatGPT Ready for Public Use in Organ-Specific Drug Toxicity Research?	2025	Drug Discovery Today	10.1016/j.drudis.2025.104297	Eligible field (≥3 studies): pharmacology + Q1 journal

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Table 8 (*continued*).

Title	Year	Venue	DOI	Reason for inclusion
Leveraging FDA Labeling Documents and Large Language Model to Enhance Annotation, Profiling, and Classification of Drug Adverse Events with AskFDALabel	2025	Drug Safety	10.1007/s40264-025-01520-1	Eligible field (≥3 studies): pharmacology + Q1 journal
BiSpec Pairwise AI: guiding the selection of bispecific antibody target combinations with pairwise learning and GPT augmentation	2024	Journal of Cancer Research and Clinical Oncology	10.1007/s00432-024-05740-3	Eligible field (≥3 studies): pharmacology + Q1 journal
Retrieval-Augmented Generation for Large Language Models in Radiology: Another Leap Forward in Board Examination Performance	2024	Radiology	10.1148/radiol.241489	Eligible field (≥3 studies): reporting + Q1 journal
Retrieval-augmented generation improves precision and trust of a GPT-4 model for emergency radiology diagnosis and classification: a proof-of-concept study	2025	European Radiology	10.1007/s00330-025-11445-z	Eligible field (≥3 studies): reporting + Q1 journal
Exploring the clinical translation of generative models like ChatGPT: promise and pitfalls in radiology, from patients to population health	2023	Journal of the American College of Radiology	10.1016/j.jacr.2023.07.007	Eligible field (≥3 studies): reporting + Q1 journal
Report Generation from X-ray imaging by Retrieval-Augmented Generation and improved Image-Text Matching	2024	IJCNN	10.1109/IJCNN60899.2024.10650332	Eligible field (≥3 studies): reporting + CORE B conference
BioinspiredLLM: Conversational Large Language Model for the Mechanics of Biological and Bio-Inspired Materials	2023	Advanced Science	10.1002/advs.202306724	Eligible field (≥3 studies): scientific research + Q1 journal
RefAI: a GPT-powered retrieval-augmented generative tool for biomedical literature recommendation and summarization	2024	Journal of the American Medical Informatics Association	10.1093/jamia/ocae129	Eligible field (≥3 studies): scientific research + Q1 journal
Large Language Models, scientific knowledge and factuality: A framework to streamline human expert evaluation	2024	Journal of Biomedical Informatics	10.1016/j.jbi.2024.104724	Eligible field (≥3 studies): scientific research + Q1 journal
LmRaC: a functionally extensible tool for LLM interrogation of user experimental results	2024	Bioinformatics	10.1093/bioinformatics/btae679	Eligible field (≥3 studies): scientific research + Q1 journal
Augmenting Large Language Models With Automated, Bibliometrics-Powered Literature Search for Knowledge Distillation: A Pilot Study for Common Spinal Pathologies	2025	Neurosurgery	10.1227/neu.0000000000003354	Eligible field (≥3 studies): scientific research + Q1 journal

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.engappai.2026.115912.

Data availability

No data was used for the research described in the article.

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